PARTE 2

RADIACIÓN SOLAR SOBRE SUPERFICIES

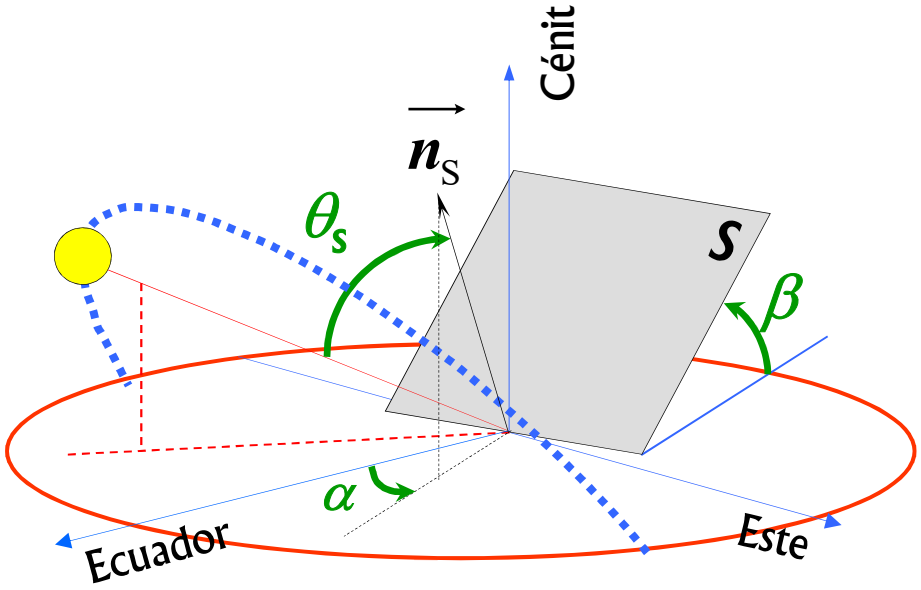
INDICE

1. Irradiación diaria

2. Potencial de irradiación anual

1. IRRADIACIÓN DIARIA SOBRE UNA SUPERFICIE

Caracterización de una superficie captadora de la energía solar



α : acimut (orientación)

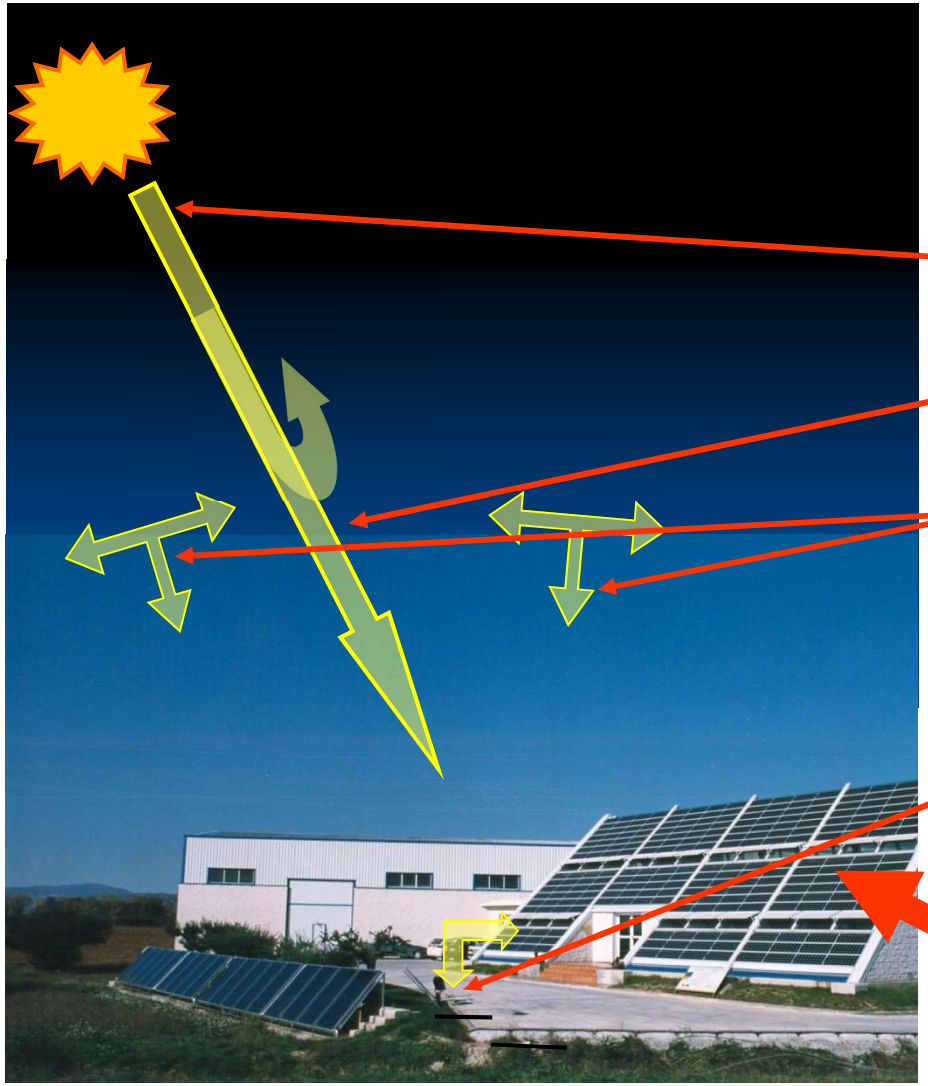
β : inclinación

θ_s : ángulo de incidencia solar

→ Determina el aprovechamiento del recurso solar

Recurso solar sobre una superficie

Definiciones



IRRADIANCIA: Densidad de Potencia (W/m^2)

IRRADIACIÓN: Densidad de Energía (Wh/m^2)

- **Extraatmosférica:** Fuera de la atmósfera. $B_0 = 1367\ W/m^2$
- **Directa:** Procede del sol y depende de su posición
- **Difusa:** Procede de la atmósfera y es la consecuencia de los procesos de reflexión, difracción, dispersión y absorción
- **Reflejada:** Procede de la reflexión de la radiación incidente sobre el entorno

Global = Directa + Difusa + Reflejada

Nomenclatura

- Extraatmosférica (B_O)
- Directa (B)
- Difusa (D)
- Reflejada (R)
- Global (G)

Subíndices:

—		irradiancia
1°:	h	irradiación horaria
	a	irradiación anual
	d	irradiación diaria
2°:	m	promedio mensual
	a	promedio anual

Ejemplos:

- $G(\alpha,\beta)$ → Irradiancia global sobre superficie con ubicación (α,β)
  $G(0) = G(0,0)$ → Irradiancia global sobre superficie horizontal
- $G_h(\alpha,\beta)$ → Irradiación global horaria
  $G_d(\alpha,\beta)$ → Irradiación global diaria
  $G_a(\alpha,\beta)$ → Irradiación global anual
- $G_{dm}(\alpha,\beta)$ → Promedio mensual de irradiación global diaria
  $G_{da}(\alpha,\beta)$ → Promedio anual de irradiación global diaria

MÉTODOS BASADOS EN PROMEDIOS DIARIOS MENSUALES

Base de cálculo diaria (días típicos):

1. Datos de partida: irradiación global diaria sobre superficie horizontal
2. Cálculo de las componentes sobre superficie horizontal
3. Descomposición de las componentes diarias en valores horarios
4. Cálculo de las componentes horarias sobre superficie inclinada → irradiación global diaria sobre superficie inclinada

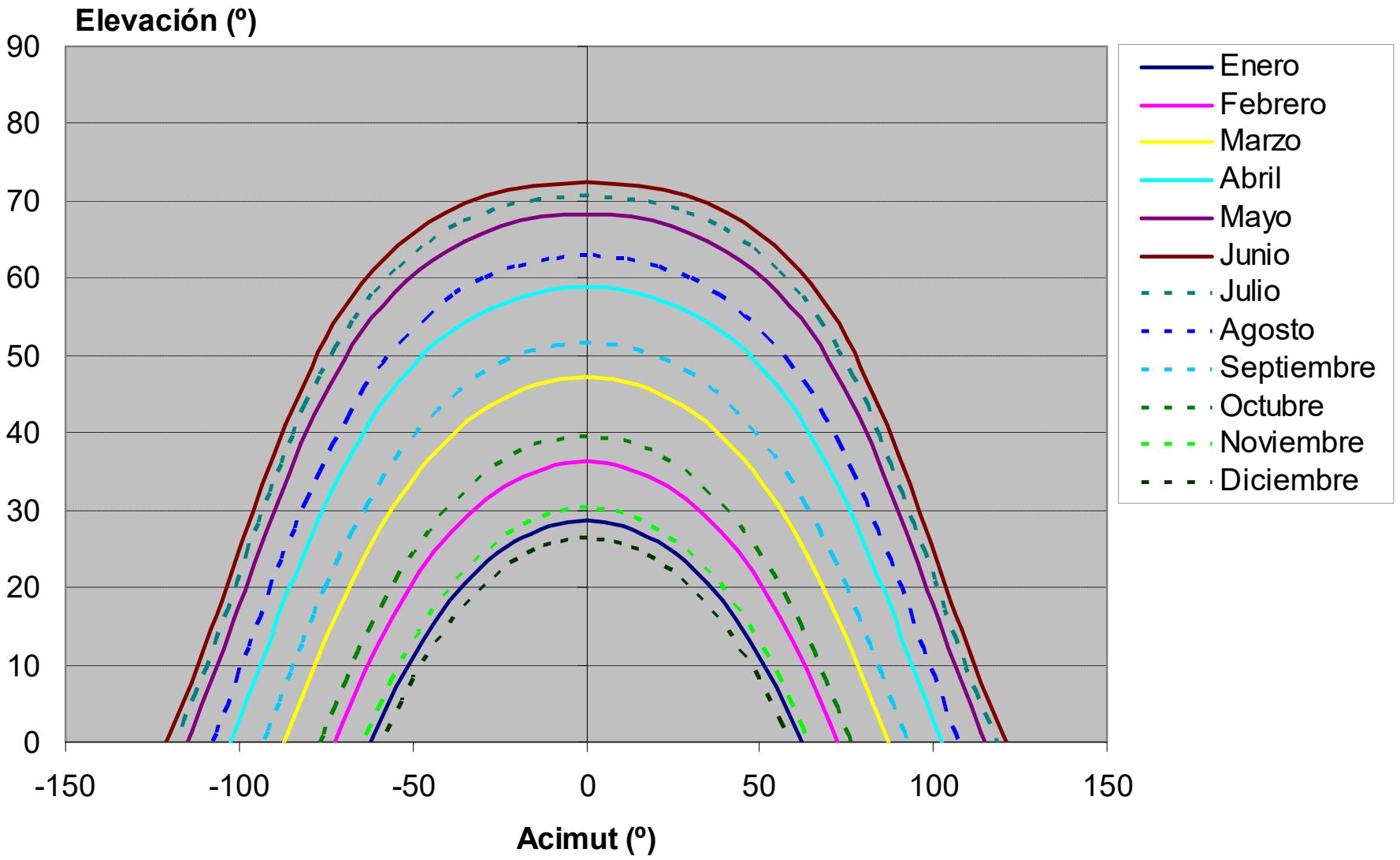
Cálculos a realizar en los días típicos de cada mes

(declinación = promedio mensual)

Mes	Dia juliano (d_n)	Mes	Dia juliano
1	17	7	198
2	46	8	228
3	75	9	258
4	105	10	289
5	135	11	319
6	161	12	345

Hoja de cálculo de apoyo: "Calculos_RadiacionSolar_IES-UPM_2024"

Trayectorias del sol en los días típicos (Madrid)



1.1. DATOS DE PARTIDA

IRRADIACIÓN GLOBAL SOBRE SUPERFICIE HORIZONTAL

- Medidas directas





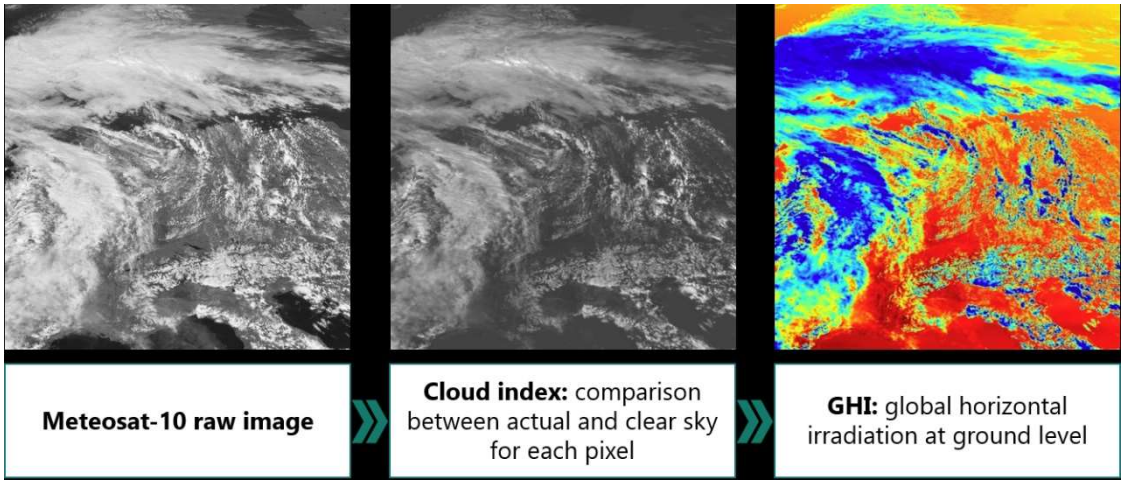


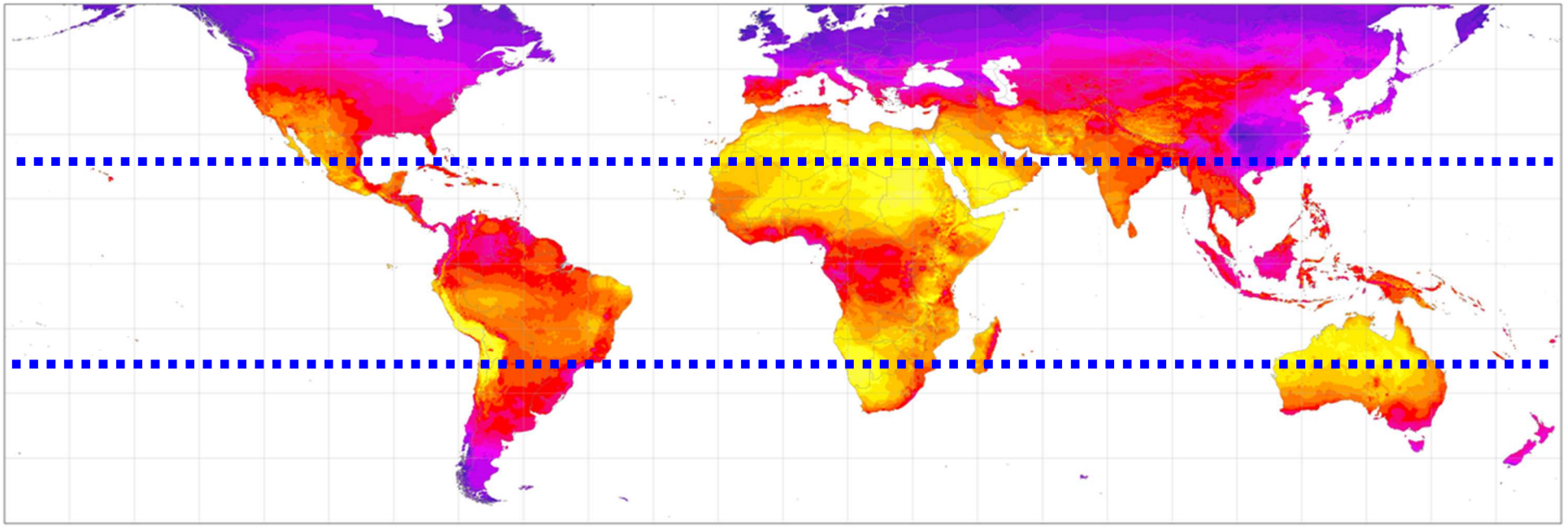
- Medidas indirectas

Horas de sol

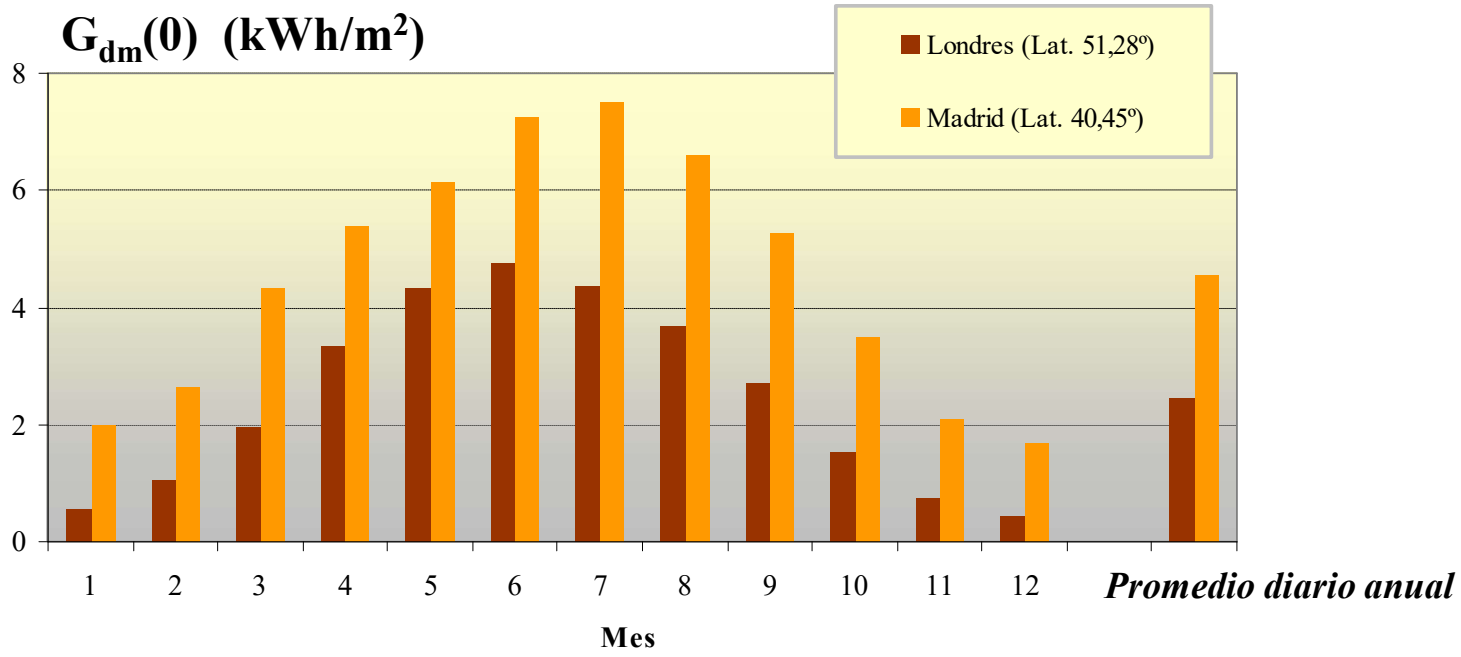


Satélite

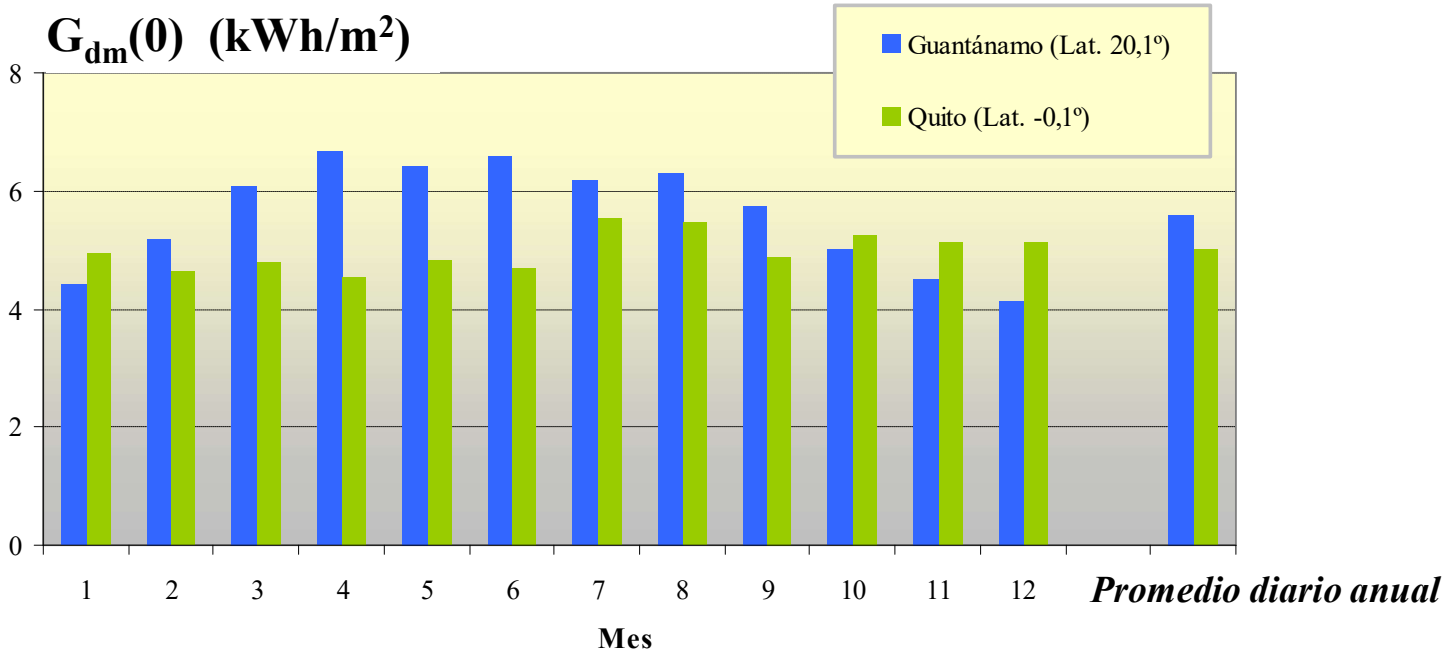




Ejemplos:



Promedio diario anual

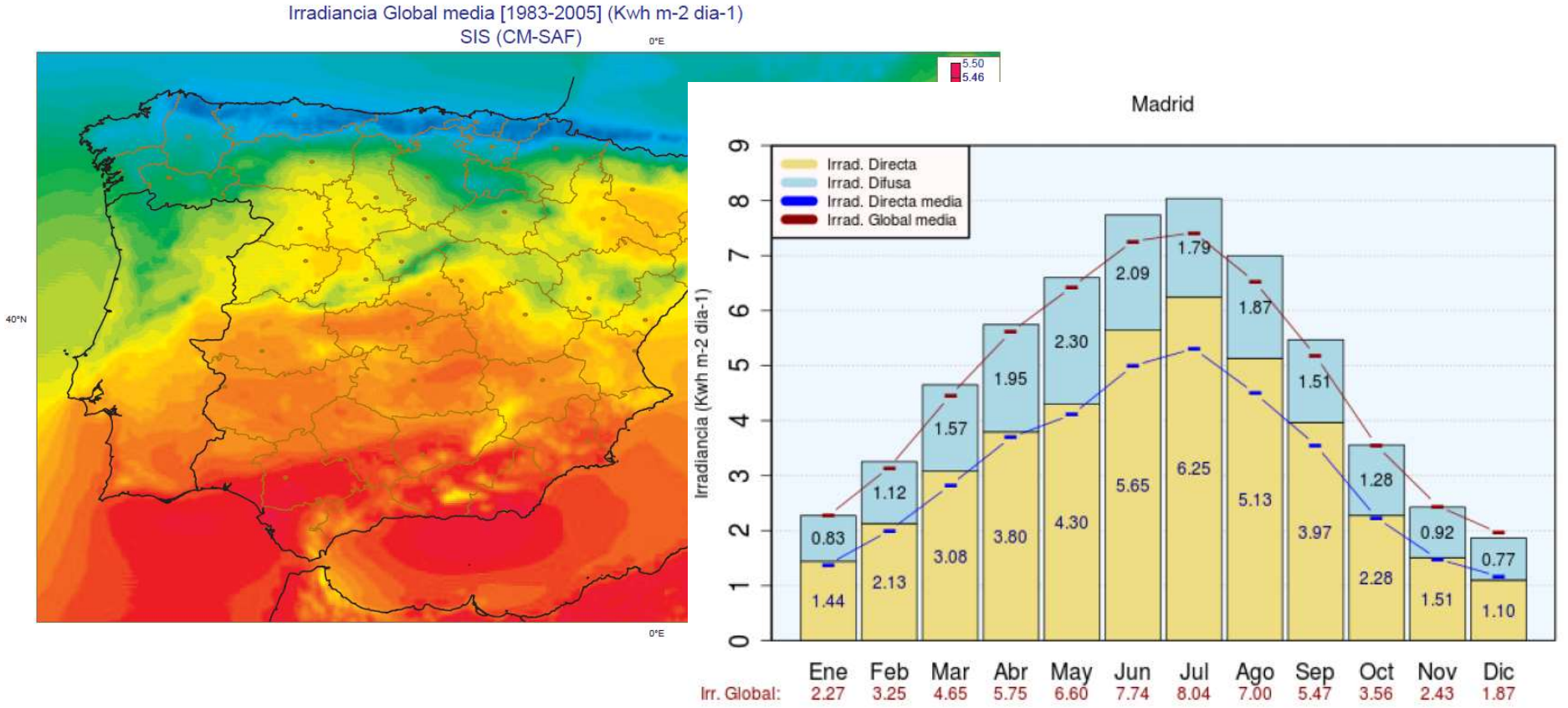


Promedio diario anual

RECURSOS EN LA RED

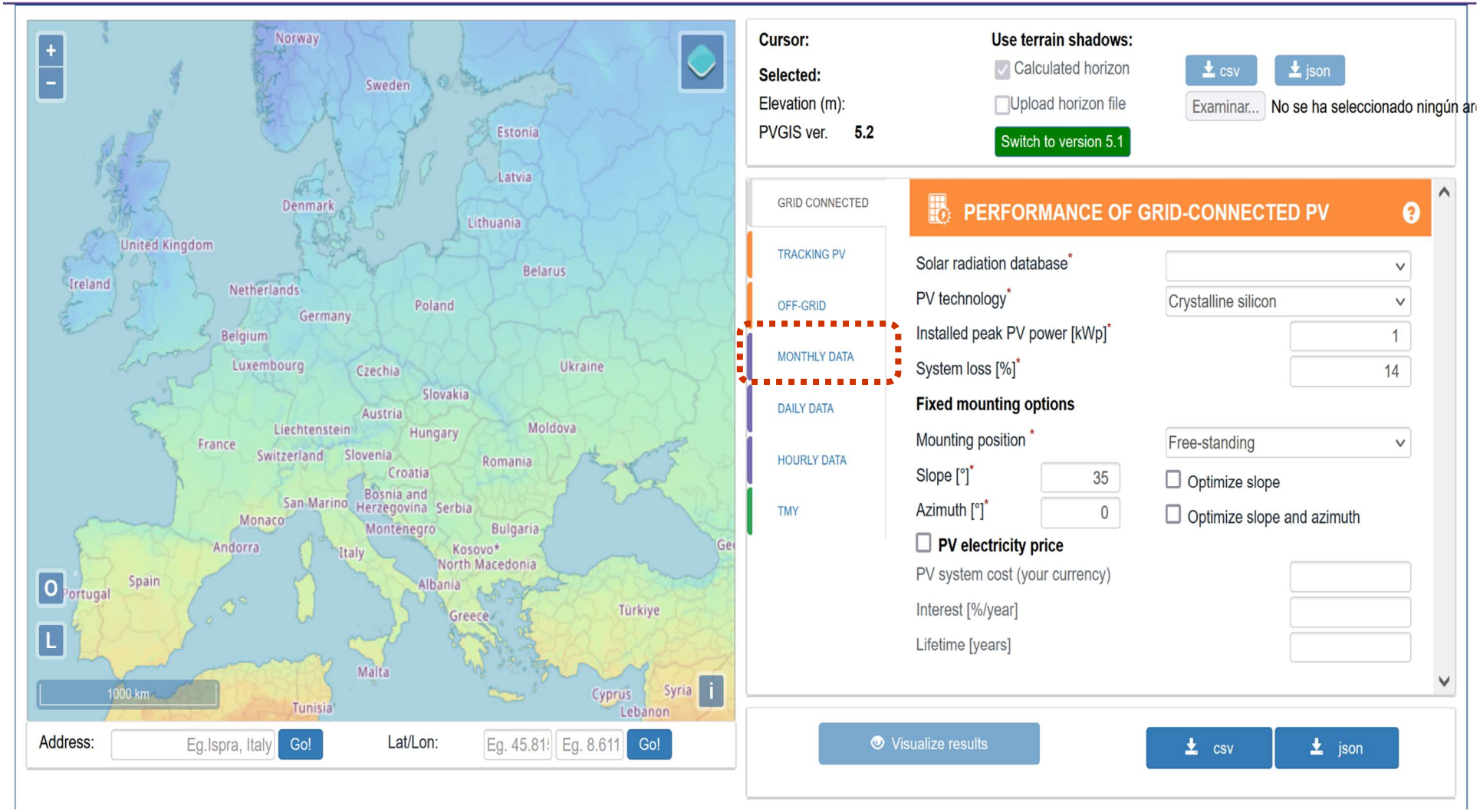
España: Atlas de Radiación Solar (AEMET, 2012)

https://www.aemet.es/documentos/es/serviciosclimaticos/datosclimatologicos/atlas_radiacion_solar/atlas_de_radiacion_24042012.pdf

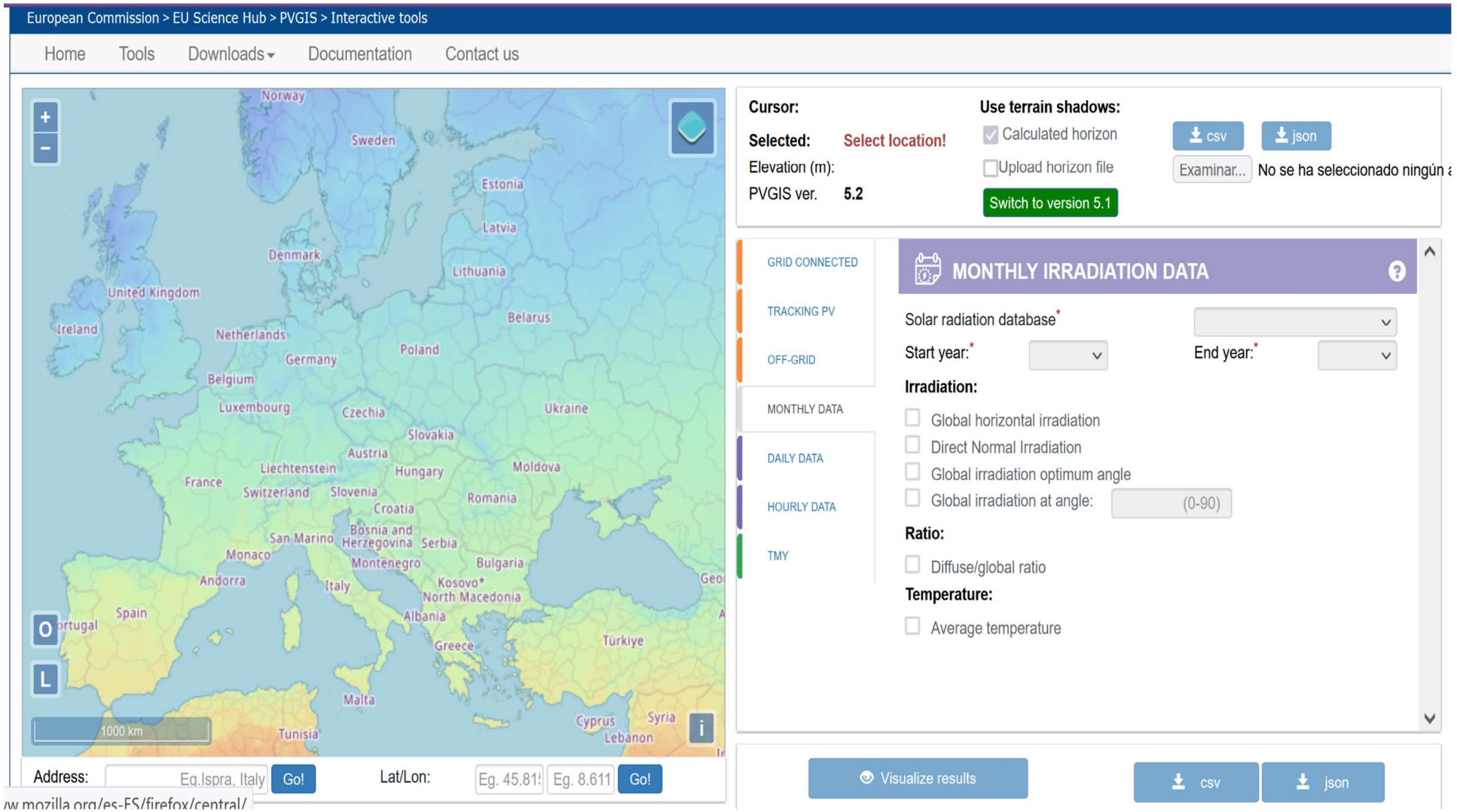


PVGIS: Photovoltaic Geographical Information System

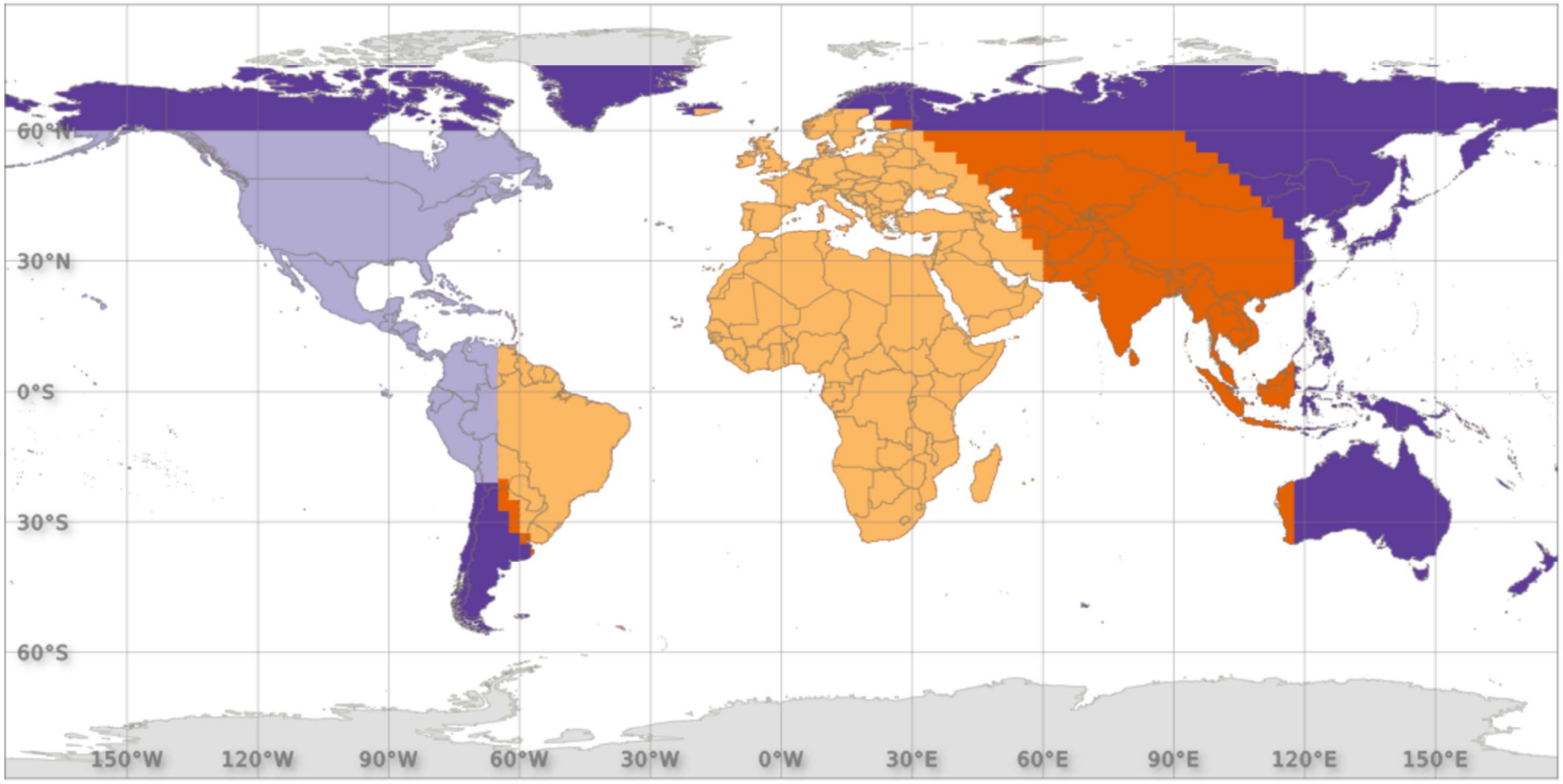
http://re.jrc.ec.europa.eu/pvgis.html



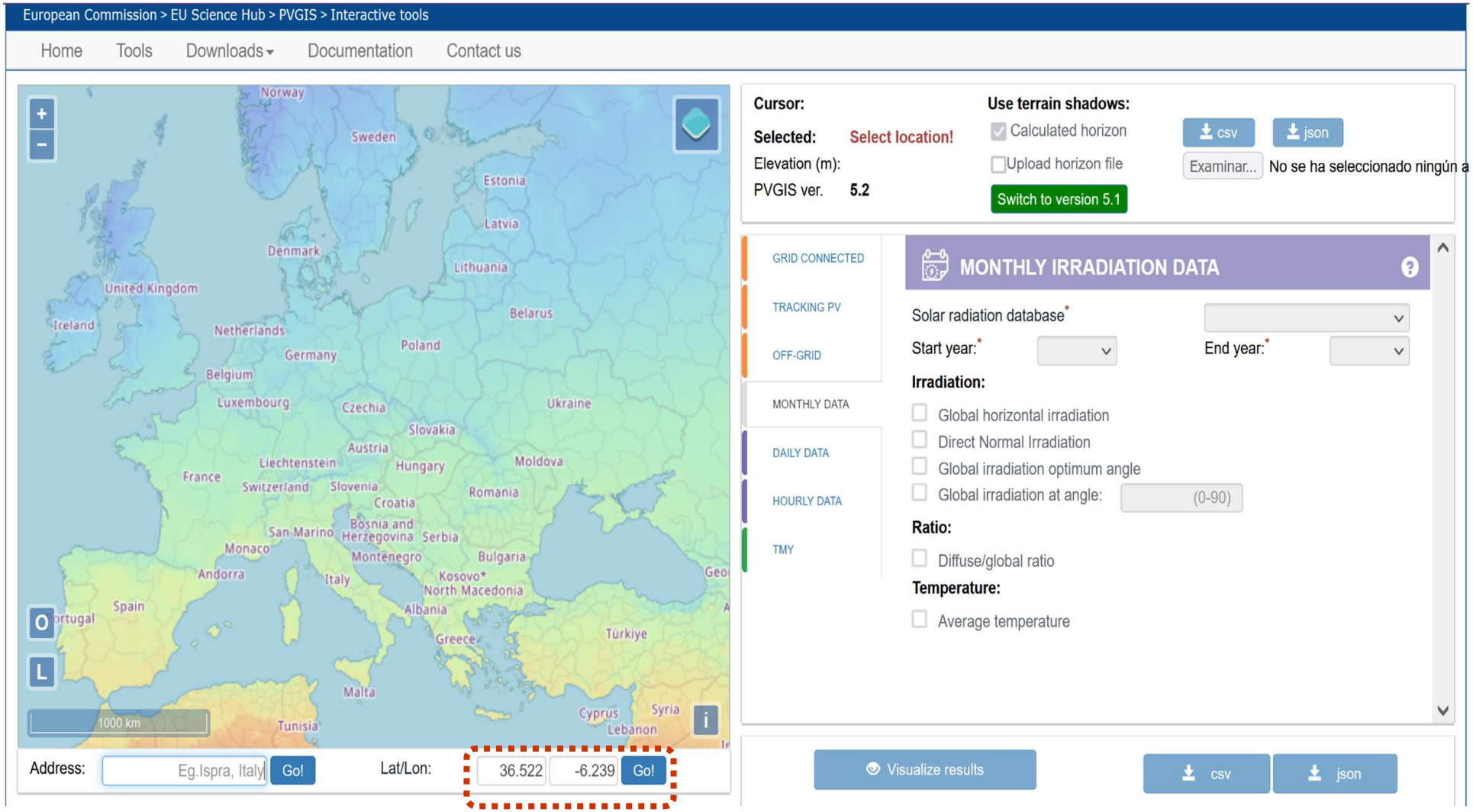
Ejemplos:

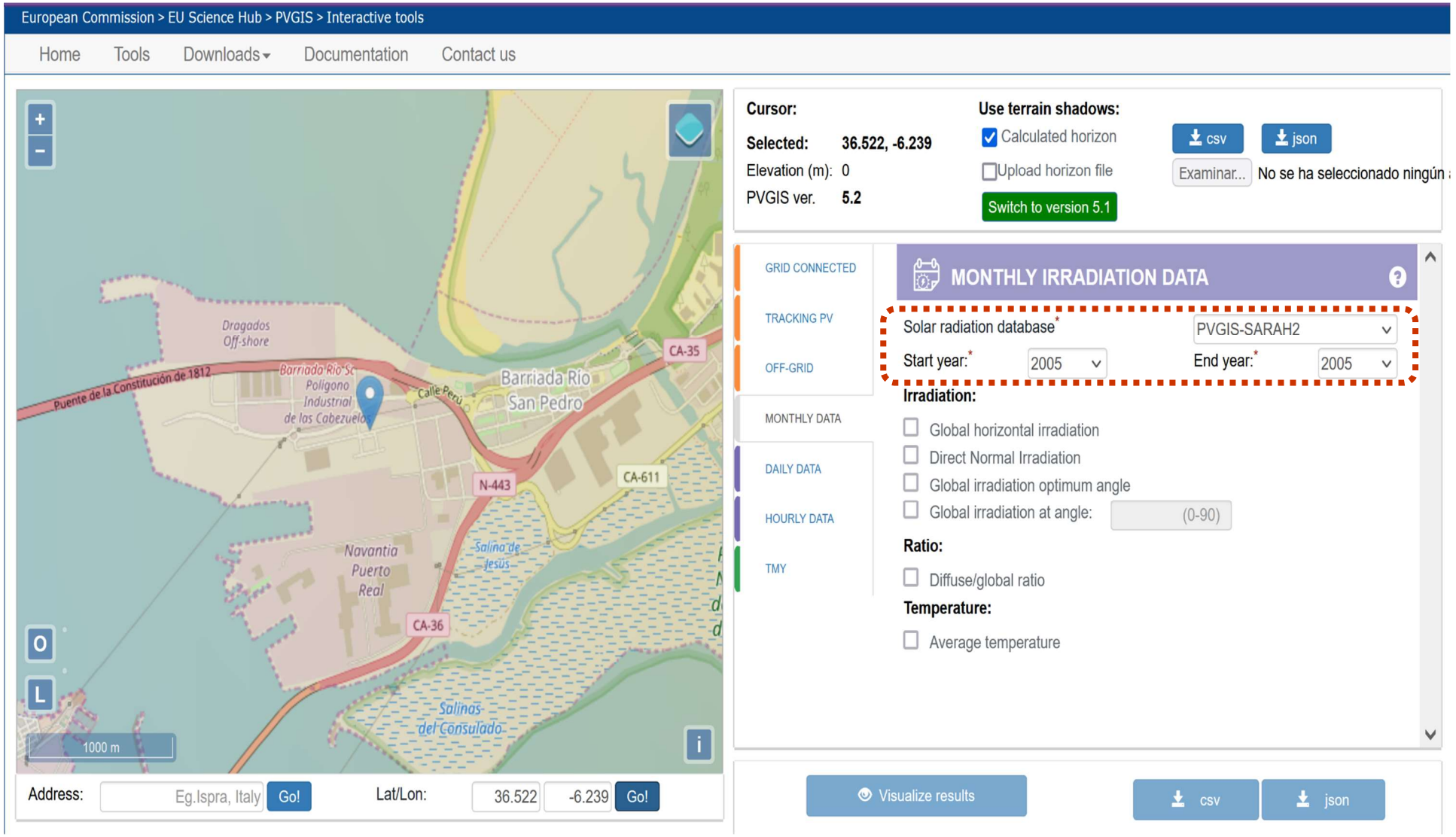

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Romania
Croatia
San Marino
Bosnia and Herzegovina
Serbia
Monaco
Montenegro
Bulgaria
Andorra
Italy
Kosovo*
North Macedonia
Spain
Albania
Greece
Türkiye
Malta
Cyprus
Syria
Lebanon
Tunisia
1000 km
Address:
Eg.Ispra, Italy
Go!
Lat/Lon:
Eg. 45.81
Eg. 8.611
Go!
Cursor:
Selected:
Select location!
Elevation (m):
PVGIS ver.
5.2
Use terrain shadows:
Calculated horizon
Upload horizon file
Switch to version 5.1
csv
json
Examinar...
No se ha seleccionado ningún a
GRID CONNECTED
TRACKING PV
OFF-GRID
MONTHLY DATA
DAILY DATA
HOURLY DATA
TMY
MONTHLY IRRADIATION DATA
Solar radiation database*
Start year:*
End year:*
Irradiation:
Global horizontal irradiation
Direct Normal Irradiation
Global irradiation optimum angle
Global irradiation at angle:
(0-90)
Ratio:
Diffuse/global ratio
Temperature:
Average temperature
Visualize results
csv
json

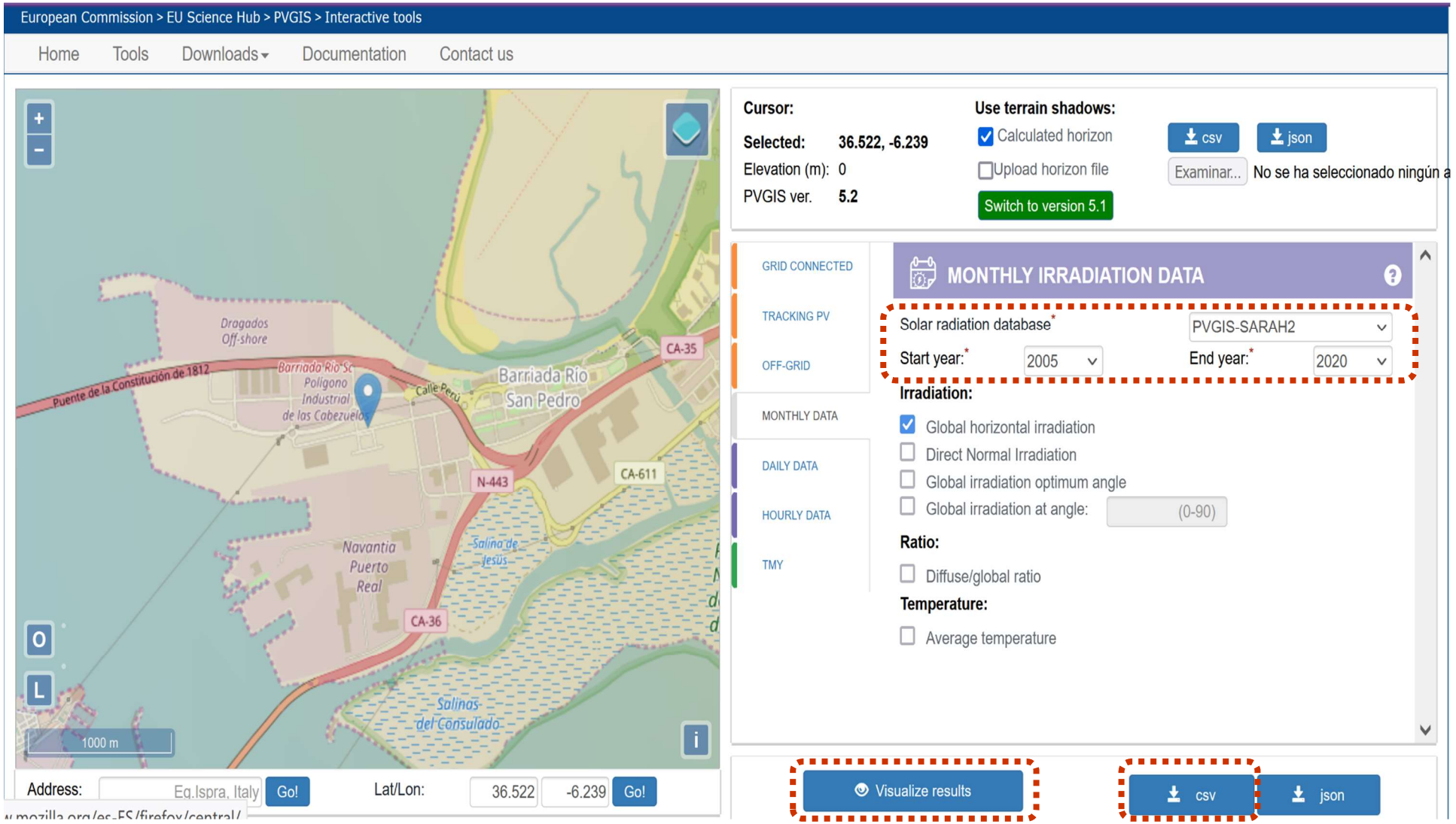
Default Solar Radiation Databases



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Selected:
Select location!
Elevation (m):
PVGIS ver.
5.2
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Calculated horizon
Upload horizon file
Switch to version 5.1
csv
json
Examinar...
No se ha seleccionado ningún a
GRID CONNECTED
TRACKING PV
OFF-GRID
MONTHLY DATA
DAILY DATA
HOURLY DATA
TMY
MONTHLY IRRADIATION DATA
Solar radiation database*
Start year:*
End year:*
Irradiation:
Global horizontal irradiation
Direct Normal Irradiation
Global irradiation optimum angle
Global irradiation at angle:
(0-90)
Ratio:
Diffuse/global ratio
Temperature:
Average temperature
Address:
Eg.Ispra, Italy
Go!
Lat/Lon:
36.522
-6.239
Go!
Visualize results
csv
json
Norway
Sweden
Estonia
Latvia
Lithuania
Denmark
United Kingdom
Ireland
Netherlands
Belarus
Germany
Poland
Belgium
Luxembourg
Czechia
Ukraine
Slovakia
Austria
Liechtenstein
Hungary
Moldova
France
Switzerland
Slovenia
Romania
Croatia
San Marino
Bosnia and Herzegovina
Serbia
Monaco
Montenegro
Bulgaria
Andorra
Italy
Kosovo*
North Macedonia
Spain
Albania
Türkiye
Greece
Malta
Cyprus
Syria
Lebanon
Tunisia
1000 km


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Barriada Rio Sc
Polígono Industrial de las Cabezuelas
Calle Perú
Barriada Rio San Pedro
CA-35
N-443
CA-611
Navantia Puerto Real
Salina de Jesús
CA-36
Salinas del Consulado
1000 m
Address:
Eg.Ispra, Italy
Go!
Lat/Lon:
36.522
-6.239
Go!
Cursor:
Selected: 36.522, -6.239
Elevation (m): 0
PVGIS ver. 5.2
Use terrain shadows:
Calculated horizon
Upload horizon file
Switch to version 5.1
csv
json
Examinar...
No se ha seleccionado ningún
GRID CONNECTED
TRACKING PV
OFF-GRID
MONTHLY DATA
DAILY DATA
HOURLY DATA
TMY
MONTHLY IRRADIATION DATA
Solar radiation database*
PVGIS-SARAH2
Start year:*
2005
End year:*
2005
Irradiation:
Global horizontal irradiation
Direct Normal Irradiation
Global irradiation optimum angle
Global irradiation at angle:
(0-90)
Ratio:
Diffuse/global ratio
Temperature:
Average temperature
Visualize results
csv
json


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Salinas del Consulado
1000 m
Address:
Eg.Ispra, Italy
Go!
Lat/Lon:
36.522
-6.239
Go!
Cursor:
Selected: 36.522, -6.239
Elevation (m): 0
PVGIS ver. 5.2
Use terrain shadows:
Calculated horizon
Upload horizon file
Switch to version 5.1
csv
json
Examinar...
No se ha seleccionado ningún a
GRID CONNECTED
TRACKING PV
OFF-GRID
MONTHLY DATA
DAILY DATA
HOURLY DATA
TMY
MONTHLY IRRADIATION DATA
Solar radiation database*
PVGIS-SARAH2
Start year:*
2005
End year:*
2020
Irradiation:
Global horizontal irradiation
Direct Normal Irradiation
Global irradiation optimum angle
Global irradiation at angle:
(0-90)
Ratio:
Diffuse/global ratio
Temperature:
Average temperature
Visualize results
csv
json

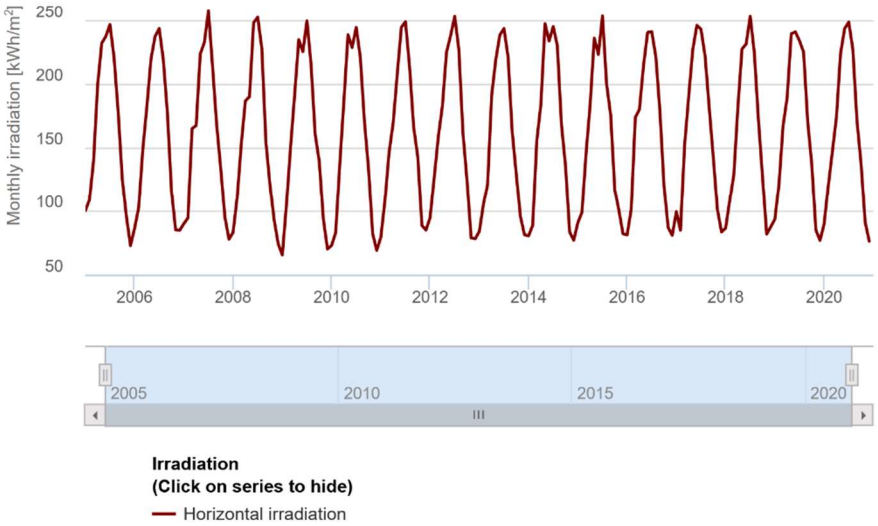
MONTHLY IRRADIATION DATA: RESULTS

Radiation | Diffuse/Glot | Temperature | Info | PDF

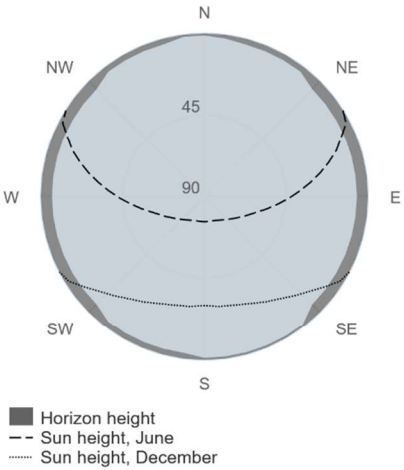
Summary

Provided inputs:	
Location [Lat/Lon]:	36.522,-6.239
Horizon:	Calculated
Database used:	PVGIS-SARAH2
Start year:	2005
End year:	2020

Monthly solar irradiation estimates



Outline of horizon



Importar datos de fichero csv:

Monthlydata_36.522_-6.239_SA2_2005_2020

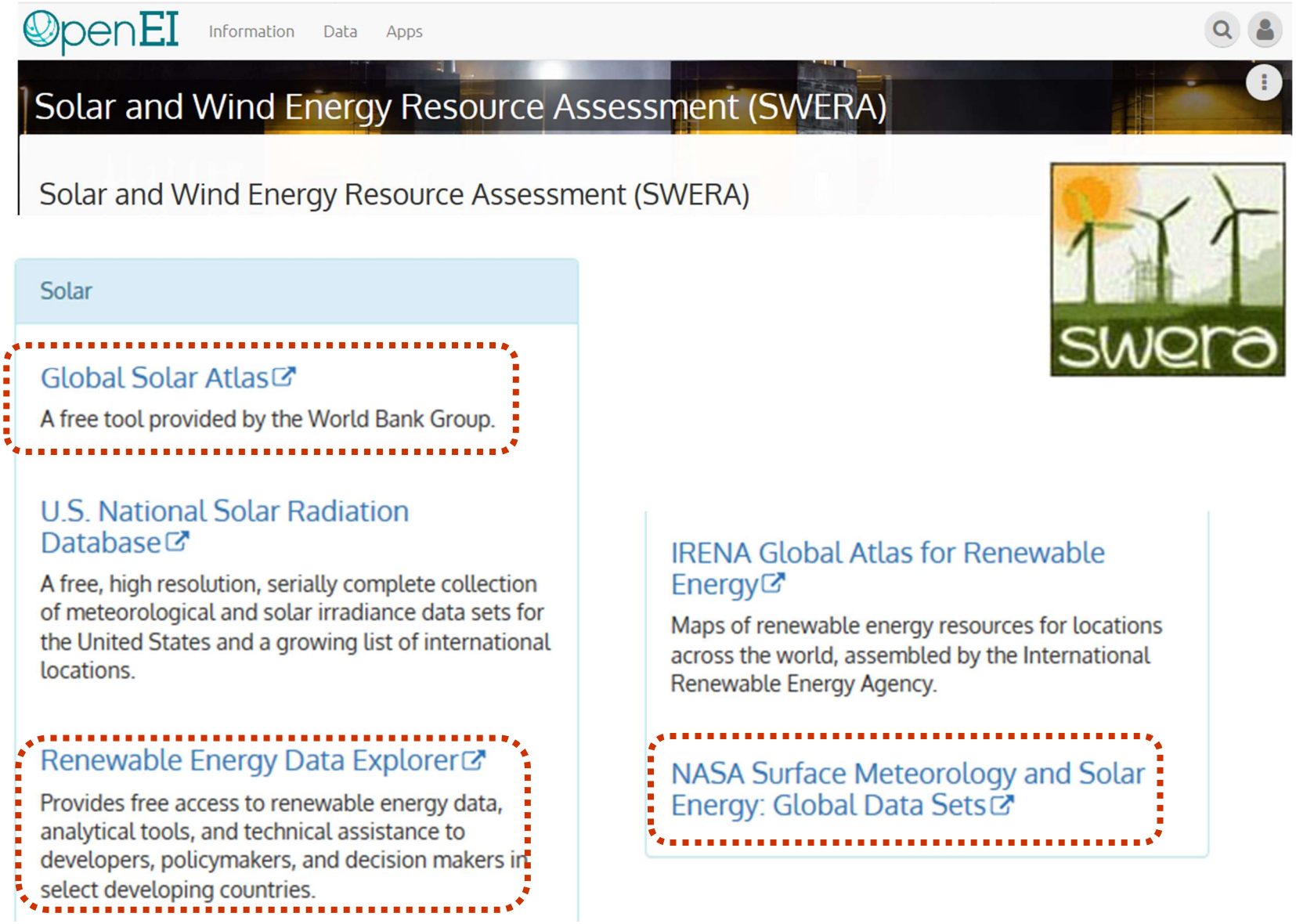
	Column1	Column2	Column3
1	Latitude	(decimal	degrees):
2	Longitude	(decimal	degrees):
3	Radiation	database:	PVGIS-SARAH2
4			
5	year	month	H(h)_m
6	2005	Jan	99.84
7	2005	Feb	108.41
8	2005	Mar	139.41
9	2005	Apr	199.15
10	2005	May	231.83
11	2005	Jun	237.24
12	2005	Jul	246.49
13	2005	Aug	220.56
14	2005	Sep	178.08
15	2005	Oct	125.34
16	2005	Nov	95.48
17	2005	Dec	72.62
18	2006	Jan	86.13
19	2006	Feb	102.29
20	2006	Mar	148.41
21	2006	Apr	182.31

Column1	Column2	Column3	Column4
Latitude	(decimal	degrees):	36.522
Longitude	(decimal	degrees):	-6.239
Radiation	database:	PVGIS-SARAH2	
year	month	H(h)_m	
2005	Jan	99.84	
2005	Feb	108.41	
2005	Mar	139.41	
2005	Apr	199.15	
2005	May	231.83	
2005	Jun	237.24	
2005	Jul	246.49	
2005	Aug	220.56	
2005	Sep	178.08	
2005	Oct	125.34	
2005	Nov	95.48	
2005	Dec	72.62	
2006	Jan	86.13	
2006	Feb	102.29	
2006	Mar	148.41	
2006	Apr	182.31	
2006	May	219.5	

Datos mensuales → pasar a promedios diarios

Solar and Wind Energy Resource Assessment (SWERA)

http://en.openei.org/apps/SWERA/



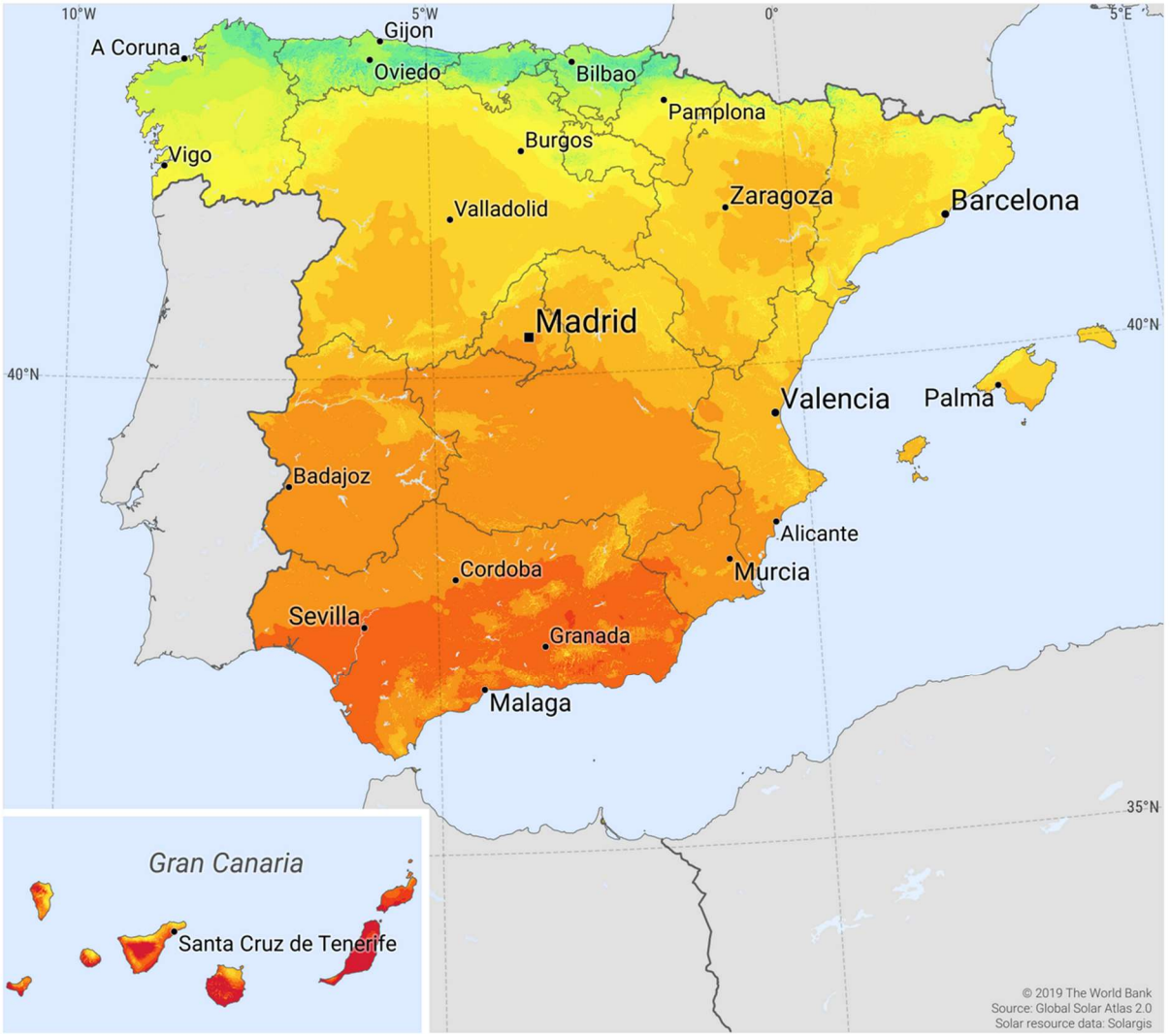
SOLAR RESOURCE MAP

GLOBAL HORIZONTAL IRRADIATION

SPAIN

WORLD BANK GROUP

ESMAP SOLARGIS



Long term average of GHI, period 1994-2018

200 km

Daily totals:	3.0	3.4	3.8	4.2	4.6	5.0	5.4	kWh/m²
Yearly totals:	1095	1241	1387	1534	1680	1826	1972	

This map is published by the World Bank Group, funded by ESMAP, and prepared by Solargis. For more information and terms of use, please visit http://globalsolaratlas.info.

NASA Prediction Of Worldwide Energy Resources

DATA ACCESS DOCUMENTATION RESOURCES ABOUT CONTACT

The POWER Project

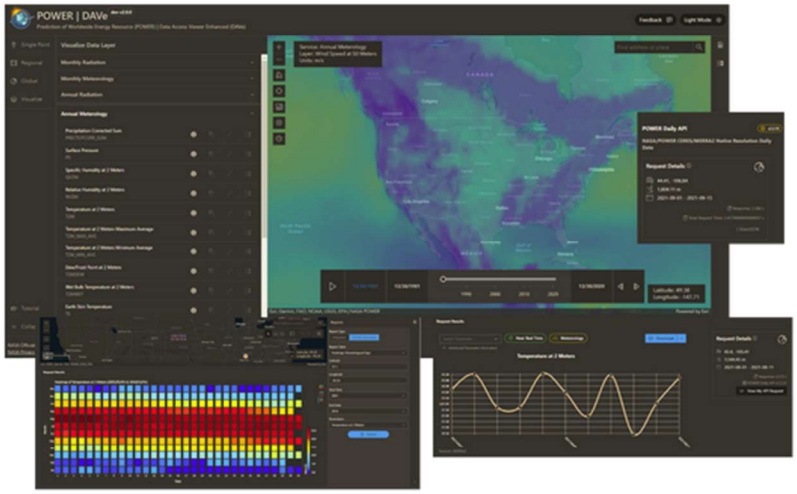
Provides solar and meteorological data sets from NASA research for support of renewable energy, building energy efficiency and agricultural needs.

Supported by NASA Earth Science's Applied Sciences Program

POWER's Web-Based Docs Pages

- Data Methodology
- Data Services Documentation
- Data Access Tutorials

POWER is hosting its **2nd Annual Virtual Global Community (GloCo) Summit event** scheduled to be held on **October 11th & 12th, 2023**. You can view registration information, the agenda, and last year's event recordings here: 2023 GloCo Event Page!



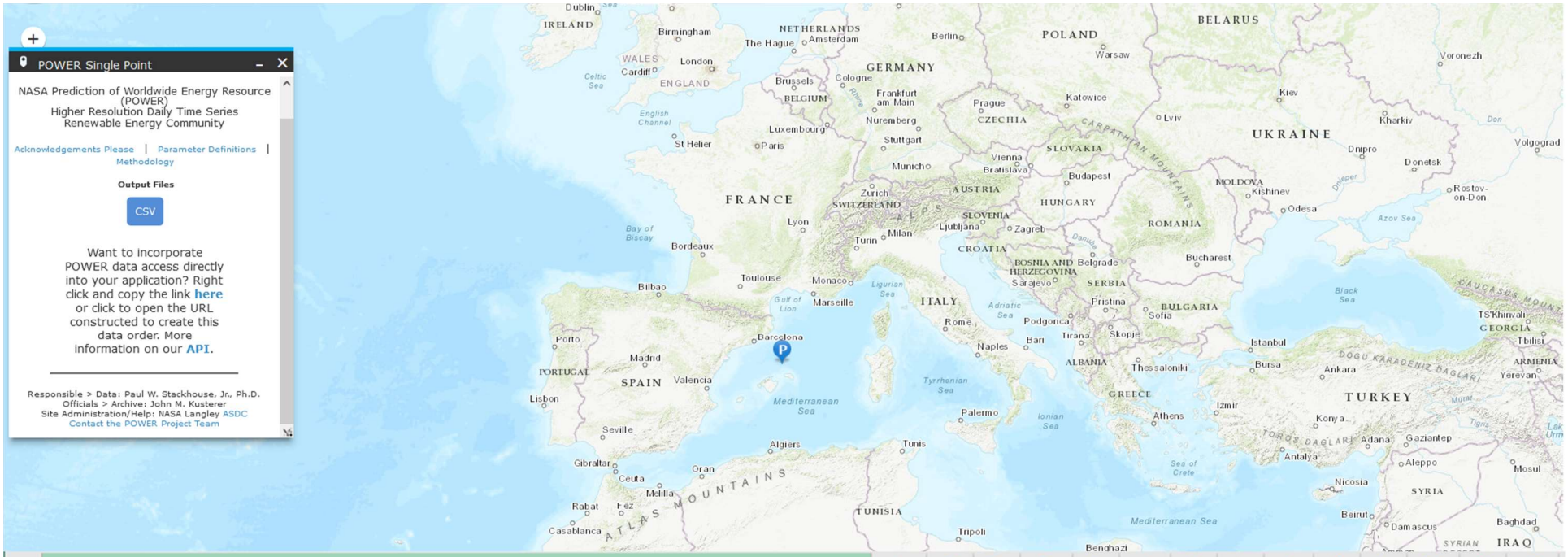
Data Access Viewer

Responsive web mapping application providing data subsetting, charting, and visualization tools in an easy-to-use interface.

DATA ACCESS VIEWER CLASSIC

ENHANCED DATA ACCESS VIEWER BETA


POWER Single Point
1. Choose a User Community
Renewable Energy
2. Choose a Temporal Average
Monthly & Annual
3. Enter Lat/Lon or Add a Point to Map
40.45
(-90 to +90 decimal degrees)
Clear
3.71
(-180 to +180 decimal degrees)
4. Select Time Extent
Start Date
2013
(YYYY)
End Date
2022
(YYYY)
5. Select Output File Format
CSV
6. Select Parameters
(Limit 20 parameters)
The Climatology temporal period has the most parameters.
Calgary
Regina
Winnipeg
Lake Nipigon
Lake Superior
Lake Michigan
Lake Huron
Lake Ontario
Lake Erie
Gulf of St. Lawrence
Quebec
Montreal
Ottawa
Toronto
Halifax
NEW BRUNSWICK
NOVA SCOTIA
MAINE
Boston
Providence
New York
Philadelphia
Washington
Albany
Buffalo
Rochester
Milwaukee
Grand Rapids
Detroit
Chicago
Cleveland
Pittsburgh
Columbus
Indianapolis
Cincinnati
Louisville
Richmond
Norfolk
Nashville
Knoxville
Greensboro
Raleigh
Charlotte
Memphis
Greenville
Atlanta
Birmingham
Jacksonville
Orlando
Tampa
Miami
New Orleans
Houston
San Antonio
Austin
Dallas
Oklahoma City
Kansas City
St Louis
Minneapolis
Denver
Salt Lake City
Las Vegas
Los Angeles
San Diego
Tijuana
Phoenix
Tucson
El Paso
Hermosillo
Chihuahua
Culiacan
Torreón
Monterrey
San Luis Potosí
Havana
Gulf of Mexico
MÉXICO
UNITED STATES
GREAT PLAINS
ROCKY MOUNTAINS
APPALACHIAN MOUNTAINS
PIEDMONT
COASTAL PLAIN
GREAT BASIN
COLORADO PLATEAU
NORTH DAKOTA
SOUTH DAKOTA
MONTANA
WYOMING
NEBRASKA
KANSAS
COLORADO
NEVADA
UTAH
ARIZONA
NEW MEXICO
OKLAHOMA
TEXAS
MINNESOTA
IOWA
MISSOURI
ARKANSAS
LOUISIANA
WISCONSIN
ILLINOIS
MICHIGAN
INDIANA
OHIO
KENTUCKY
TENNESSEE
MISSISSIPPI
ALABAMA
GEORGIA
FLORIDA
SOUTH CAROLINA
NORTH CAROLINA
VIRGINIA
6. Select Parameters (Limit 20 parameters)
The Climatology temporal period has the most parameters.
Double-click folders to expand and show available parameters.
Search Parameters
Solar Fluxes and Related
Parameters for Tilted PV Panels
All Sky Surface Shortwave Downward Irradiance
Clear Sky Surface Shortwave Downward Irradiance
All Sky Surface Albedo
All Sky Insolation Clearness Index
Parameters for Solar Cooking
Temperature/Thermal IR Flux
Humidity/Precipitation
Wind/Pressure
Parameter Definitions | Methodology



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	-BEGIN HEADER-														
2	NASA/POWER CERES/MERRA2 Native Resolution Monthly and Annual														
3	Dates (month/day/year): 01/01/2013 through 12/31/2022														
4	Location: Latitude 40.45 Longitude 3.71														
5	Elevation from MERRA-2: Average for 0.5 x 0.625 degree lat/lon region = 1.7 meters														
6	The value for missing source data that cannot be computed or is outside of the sources availability range: -999														
7	Parameter(s):														
8	ALLSKY_SFC_SW_DWN CERES SYN1deg All Sky Surface Shortwave Downward Irradiance (kW-hr/m^2/day)														
9	-END HEADER-														
10	PARAMETER	YEAR	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC	ANN
11	ALLSKY_SFC_SW_DWN	2013	2.1	2.7	4.24	5.3	6.49	7.7	7.39	6.28	4.86	3.23	2.04	1.88	4.53
12	ALLSKY_SFC_SW_DWN	2014	1.94	3.25	4.33	5.95	6.71	7.56	7.14	6.04	4.65	3.81	2.17	1.71	4.61
13	ALLSKY_SFC_SW_DWN	2015	2.16	2.75	4.04	5.86	6.99	7.74	7.47	6.23	4.55	3.3	2.57	1.95	4.64
14	ALLSKY_SFC_SW_DWN	2016	2.1	2.93	4.31	5.19	6.55	7.27	7.47	6.66	5.01	3.27	2.23	2.02	4.59
15	ALLSKY_SFC_SW_DWN	2017	2.0	3.11	4.94	6.18	7.15	7.74	7.3	6.2	4.58	3.53	2.47	1.66	4.75
16	ALLSKY_SFC_SW_DWN	2018	2.22	2.35	4.44	5.58	6.26	7.2	7.54	6.25	4.67	3.28	2.16	2.02	4.51
17	ALLSKY_SFC_SW_DWN	2019	2.22	3.58	4.98	5.42	6.73	7.62	7.26	6.4	4.83	3.53	2.37	1.91	4.74
18	ALLSKY_SFC_SW_DWN	2020	2.11	3.32	3.95	5.35	6.93	7.35	7.23	6.4	4.96	3.55	2.34	1.84	4.61
19	ALLSKY_SFC_SW_DWN	2021	2.08	2.8	4.35	5.15	6.34	7.22	7.46	6.08	4.58	3.63	1.99	1.91	4.48
20	ALLSKY_SFC_SW_DWN	2022	2.45	3.47	3.33	5.43	6.88	7.44	7.63	6.53	4.96	3.64	2.36	1.98	4.68

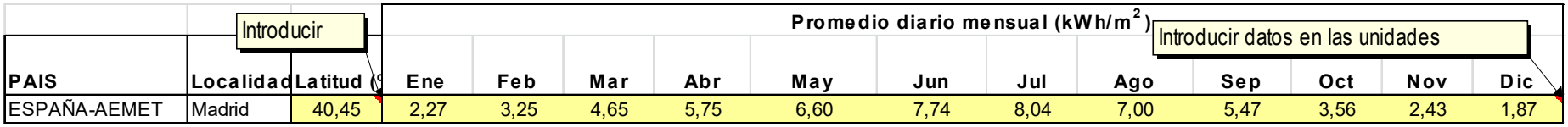
EJEMPLO

Calcular los promedios mensuales y anual de la irradiación diaria sobre una superficie orientada a sur e inclinada 38º en Madrid, utilizando la hoja de cálculo de apoyo y los datos de partida proporcionados por la AEMET.

1. Datos de partida: irradiación global diaria sobre superficie horizontal [pestaña "Gdm"]

Fuente: AEMET, Atlas de Radiación Solar de España (2012)

Mes	$G_{dm}(0)$ (kWh/m²)	Mes	$G_{dm}(0)$	Mes	$G_{dm}(0)$	Mes	$G_{dm}(0)$
1	2,27	4	5,75	7	8,04	10	3,56
2	3,25	5	6,60	8	7,00	11	2,43
3	4,65	6	7,74	9	5,47	12	1,87



PAIS	Localidad	Latitud (º	Ene	Feb	Mar	Abr	May	Jun	Jul	Ago	Sep	Oct	Nov	Dic
ESPAÑA-AEMET	Madrid	40,45	2,27	3,25	4,65	5,75	6,60	7,74	8,04	7,00	5,47	3,56	2,43	1,87

2. Componentes sobre superficie horizontal: $D_{dm}(0)$, $B_{dm}(0)$ ["Gdm"]

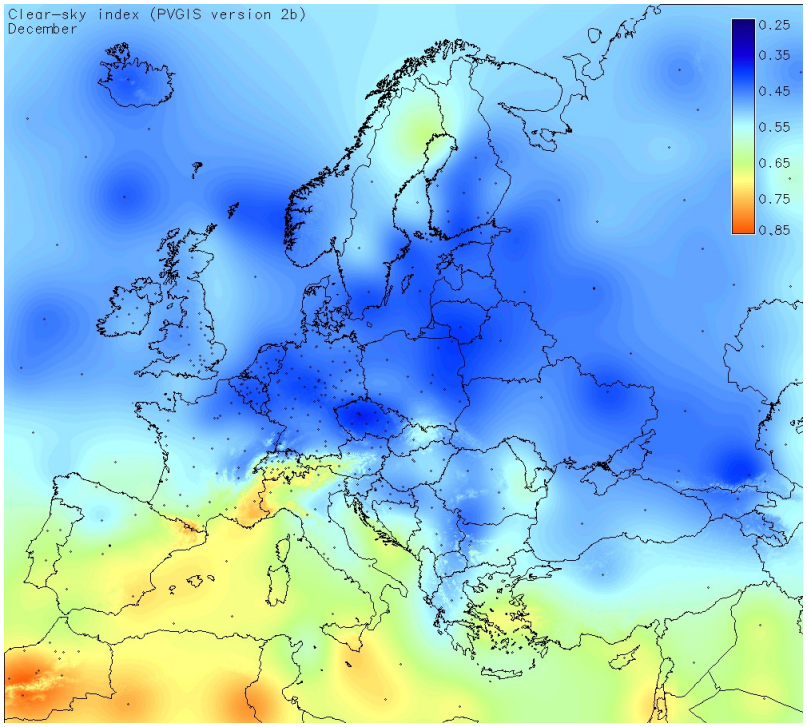
$$G_{dm}(0) = B_{dm}(0) + D_{dm}(0)$$

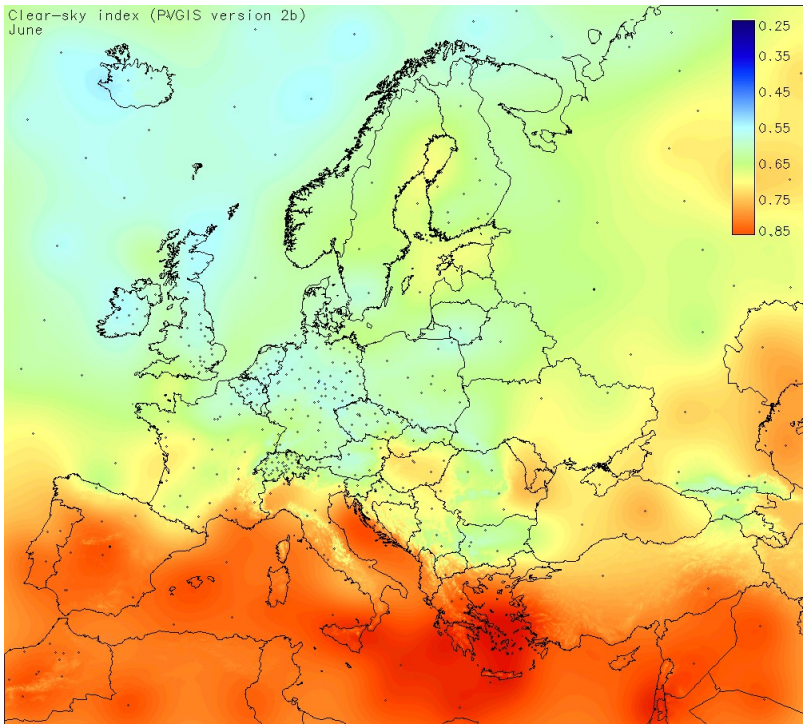
Indice de claridad (K_T):

$$K_T = \frac{\textit{Radiación global}}{\textit{Radiación extraatmosférica}} = \frac{G_{dm}(0)}{B_{0dm}(0)}$$

Fracción de difusa (K_D):

$$K_D = \frac{\textit{Radiación difusa}}{\textit{Radiación global}} = \frac{D_{dm}(0)}{G_{dm}(0)} = f(K_T, \ldots)$$

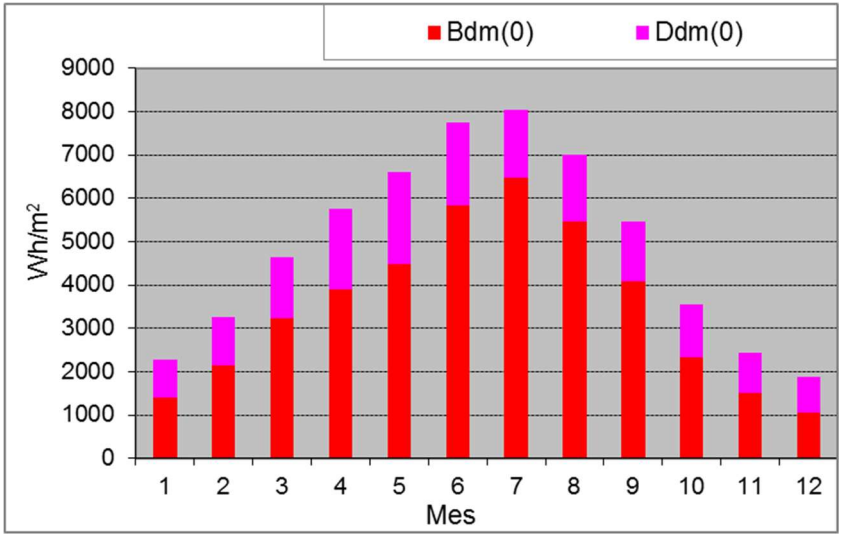




Resultado:

Latitud, Φ (º) =	*40,45*	Introducir datos
Inclinación, β (º)=	**38**	
Orientación, α (º) =	**0**	(0º: sur; 180º: norte; valores < 0: sureste/noreste; valores >0: suroeste/noroeste)
Coeficiente ρ =	**0,2**	Coeficiente de reflexión del suelo (0,2 por defecto)

		(Wh/m^2)	(Wh/m^2)	(Wh/m^2)
Mes	Dia	Gdm(0)	Ddm(0)	Bdm(0)
1	17	2270	866	1404
2	46	3250	1108	2142
3	75	4650	1420	3230
4	105	5750	1852	3898
5	135	6600	2132	4468
6	161	7740	1900	5840
7	198	8040	1572	6468
8	228	7000	1531	5469
9	258	5470	1388	4082
10	289	3560	1222	2338
11	319	2430	930	1500
12	345	1870	811	1059



3. Descomposición de componentes diarias en valores horarios ["Ghm"]

Análisis en el instante central de cada hora:

$$G_{dm}(0)=\sum G_{hm}(0) \; ; \; h = [-11,5 \; , -10,5 \; , ..., -0,5 \; , +0,5, ..., +10,5 \; , +11,5]$$

[h<0: antes del mediodía solar;
h>0: después del mediodía solar]

$$B_{dm}(0)=\sum B_{hm}(0) \; ; \; h = [-11,5 \; , -10,5 \; , ..., -0,5 \; , +0,5, ..., +10,5 \; , +11,5]$$

$$D_{dm}(0)=\sum D_{hm}(0) \; ; \; h = [-11,5 \; , -10,5 \; , ..., -0,5 \; , +0,5, ..., +10,5 \; , +11,5]$$

MES:	1
Epsilon_0 =	1,0316
Declinación (º)=	-20,92

ω (h)	D(0)	G(0)	B(0)
-11,5	0,0	0,0	0,0
-10,5	0,0	0,0	0,0
-9,5	0,0	0,0	0,0
-8,5	0,0	0,0	0,0
-7,5	0,0	0,0	0,0
-6,5	0,0	0,0	0,0
-5,5	0,0	0,0	0,0
-4,5	11,9	22,5	10,6
-3,5	59,2	132,1	72,9
-2,5	97,9	245,4	147,5
-1,5	125,2	338,3	213,1
-0,5	139,3	390,6	251,2
0,5	139,3	390,6	251,2
1,5	125,2	338,3	213,1
2,5	97,9	245,4	147,5
3,5	59,2	132,1	72,9
4,5	11,9	22,5	10,6
5,5	0,0	0,0	0,0
6,5	0,0	0,0	0,0
7,5	0,0	0,0	0,0
8,5	0,0	0,0	0,0
9,5	0,0	0,0	0,0
10,5	0,0	0,0	0,0
11,5	0,0	0,0	0,0
	867,1	**2.257,7**	**1.390,7**

4. Cálculo de las componentes horarias sobre la superficie inclinada ["Ghm"]

Modelado de la irradiación global:

$$G_{hm}(\alpha,\beta) = B_{hm}(\alpha,\beta) + D_{hm}(\alpha,\beta) + R_{hm}(\alpha,\beta)$$

- **Irradiancia Directa:** $B_{hm}(\alpha,\beta)$

- **Irradiancia Reflejada:**

$$R_{hm}(\alpha,\beta) = G_{hm}(0) \cdot \rho \cdot \frac{1-\cos\beta}{2}$$

ρ : Coeficiente de reflexión del entorno

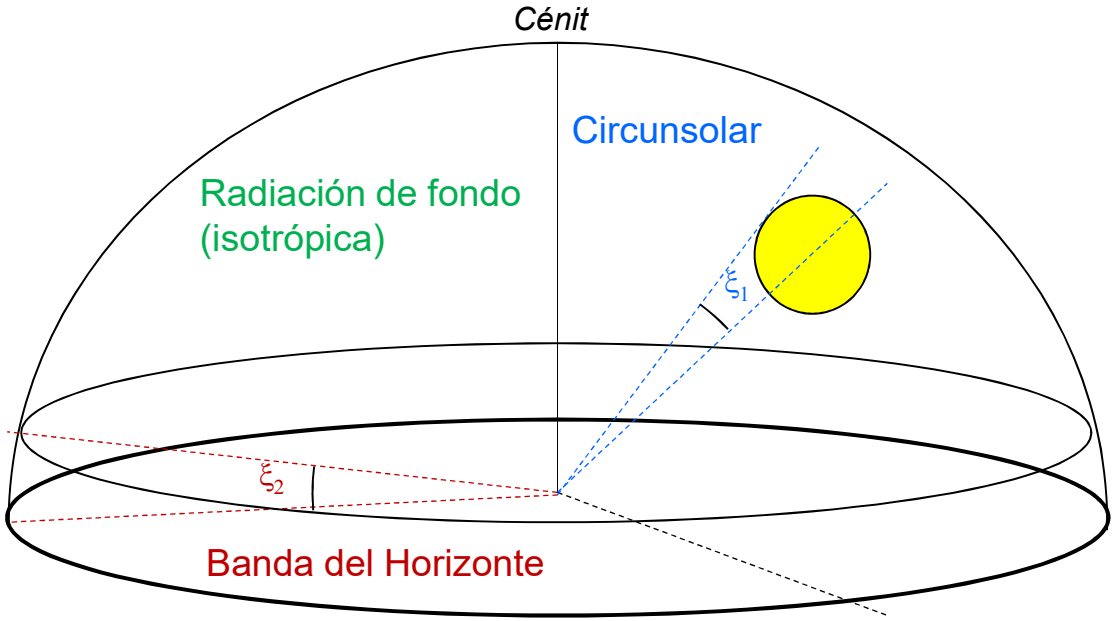
Usual values for albedo	
Urban situation	0.14 - 0.22
Grass	0.15 - 0.25
Fresh Grass	0.26
Fresh snow	0.82
Wet snow	0.55 - 0.75
Dry asphalt	0.09 - 0.15
Wet asphalt	0.18
Concrete	0.25 - 0.35
Red tiles	0.33
Aluminium	0.85
New galvanised steel	0.35
Very dirty galavanised ste	0.08

[Fuente: PVsyst]

- **Irradiación Difusa (modelo de Perez & Seals):**

$$D_{hm}(\alpha,\beta) = D_{hm}{}^{C}(\alpha,\beta) + D_{hm}{}^{BH}(\alpha,\beta) + D_{hm}{}^{I}(\alpha,\beta)$$

Circunsolar *Banda del horizonte* *Fondo*



MES:	1
Epsilon_0 =	1,0316
Declinación (°)=	-20,92

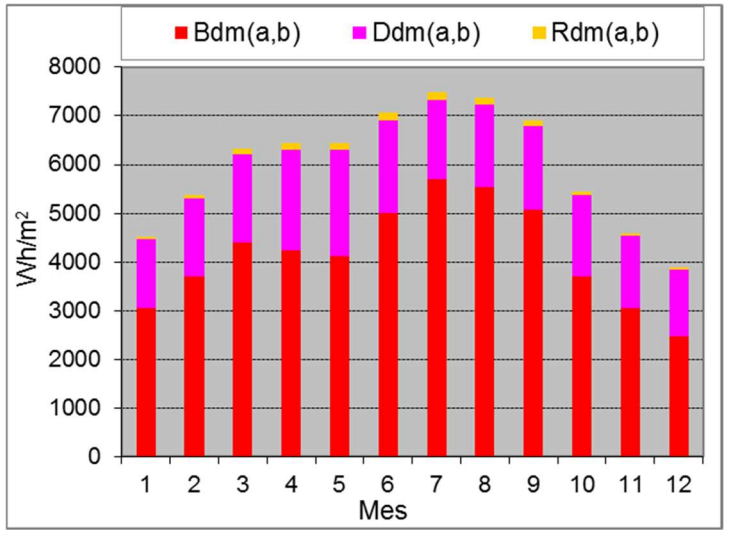
$$B_{hm}(\alpha,\beta) + D_{hm}(\alpha,\beta) + R_{hm}(\alpha,\beta) = G_{hm}(\alpha,\beta)$$

ω (h)	B(α,β)	D(α,β)	R(α,β)	G(α,β)= Ghm(α,β)
-11,5	0,0	0,0	0,0	0,0
-10,5	0,0	0,0	0,0	0,0
-9,5	0,0	0,0	0,0	0,0
-8,5	0,0	0,0	0,0	0,0
-7,5	0,0	0,0	0,0	0,0
-6,5	0,0	0,0	0,0	0,0
-5,5	0,0	0,0	0,0	0,0
-4,5	90,0	51,6	0,5	142,1
-3,5	200,5	106,7	2,8	310,0
-2,5	321,8	153,6	5,2	480,6
-1,5	424,5	189,4	7,2	621,2
-0,5	483,2	208,2	8,3	699,7
0,5	483,2	208,2	8,3	699,7
1,5	424,5	189,4	7,2	621,2
2,5	321,8	153,6	5,2	480,6
3,5	200,5	106,7	2,8	310,0
4,5	90,0	51,6	0,5	142,1
5,5	0,0	0,0	0,0	0,0
6,5	0,0	0,0	0,0	0,0
7,5	0,0	0,0	0,0	0,0
8,5	0,0	0,0	0,0	0,0
9,5	0,0	0,0	0,0	0,0
10,5	0,0	0,0	0,0	0,0
11,5	0,0	0,0	0,0	0,0
	3.040,1	**1.419,1**	**47,9**	**4.507,0**

5. Resultados: promedios diarios mensuales y anuales ["Gdm"]

Mes	Bdm(α,β) (Wh/m²)	Ddm(α,β) (Wh/m²)	Rdm(α,β) (Wh/m2)	Gdm(α,β) (Wh/m²)	
1	3040	1419	48	**4.507**	
2	3697	1597	68	**5.362**	
3	4407	1811	98	**6.316**	
4	4244	2065	121	**6.430**	
5	4130	2179	140	**6.450**	
6	4999	1895	164	**7.059**	
7	5708	1615	171	**7.494**	
8	5533	1693	148	**7.375**	
9	5080	1706	115	**6.901**	
10	3706	1666	75	**5.447**	
11	3063	1474	51	**4.588**	
12	2467	1383	39	**3.889**	
				5.987	**Promedio diario anual**

x 365



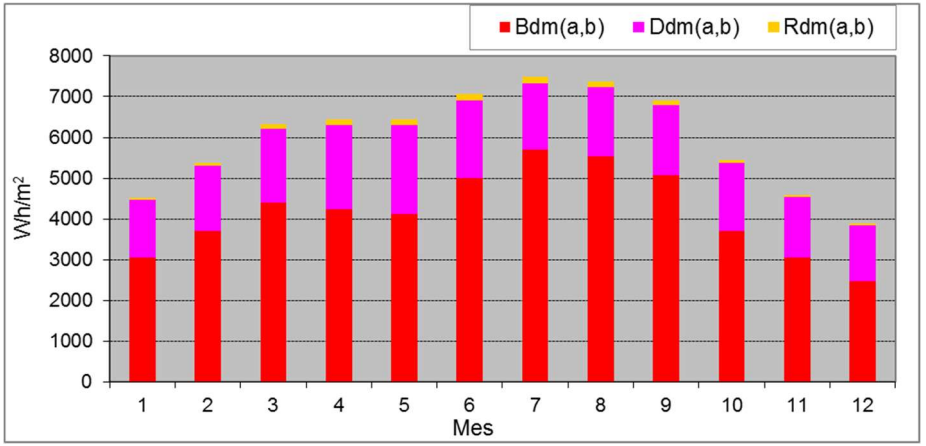
Ba(α,β) (kWh/m²)	Da(α,β) (kWh/m²)	Ra(α,β) (kWh/m²)	Ga(α,β) (kWh/m²)	
1.524	**624**	**38**	**2.185**	**Global anual**
70%	29%	2%	100%	

Ejemplos: Madrid

Latitud, Φ (º) =	40.45
Inclinación, β (º)=	38
Orientación, α (º) =	0

(Máximo ↔ 100%)

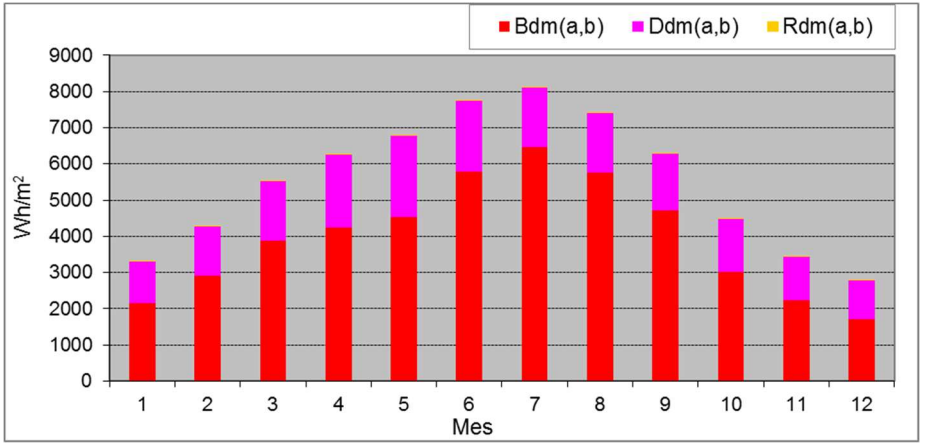
Ga(α,β) (kWh/m²)
2,185



Latitud, Φ (º) =	40.45
Inclinación, β (º)=	15
Orientación, α (º) =	0

(-7%)

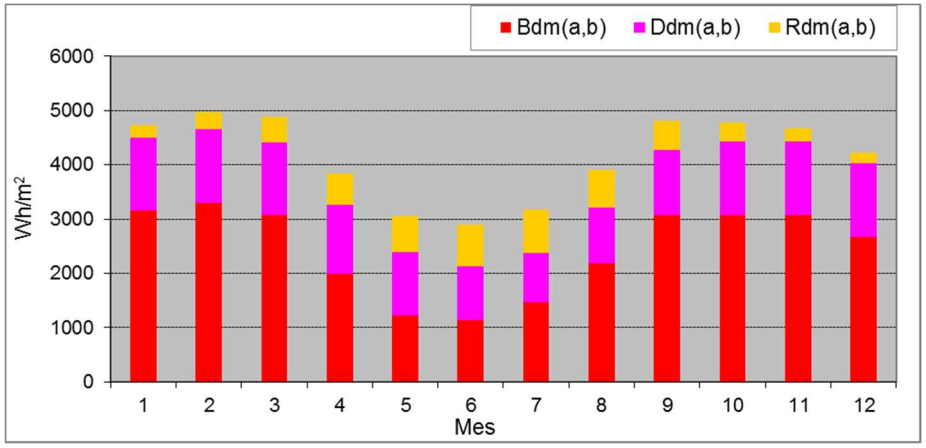
Ga(α,β) (kWh/m²)
2,026



Latitud, Φ (º) =	40.45
Inclinación, β (º)=	90
Orientación, α (º) =	0

(-30%)

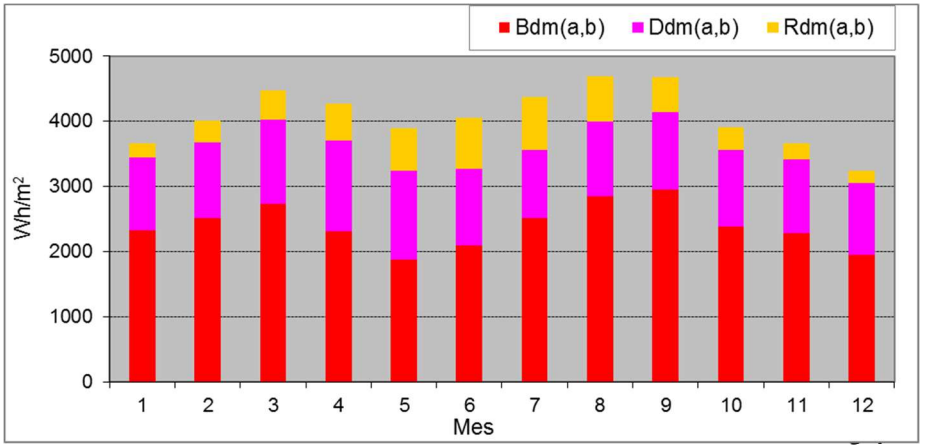
Ga(α,β) (kWh/m²)
1,517



Latitud, Φ (º) =	40.45
Inclinación, β (º)=	90
Orientación, α (º) =	45

(-32%)

Ga(α,β) (kWh/m²)
1,489



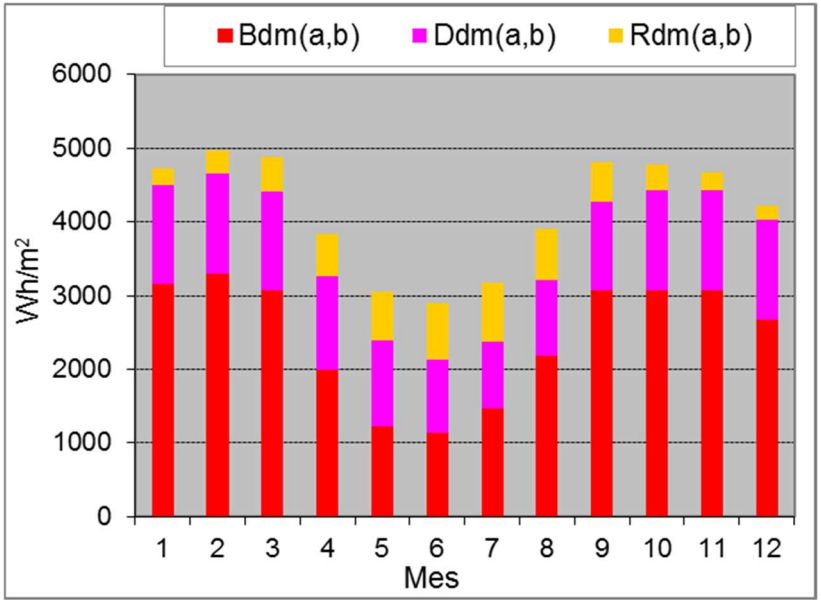
Latitud, Φ (º) =	*40,45*
Inclinación, β (º)=	**90**
Orientación, α (º) =	**0**
Coeficiente ρ =	**0,2**

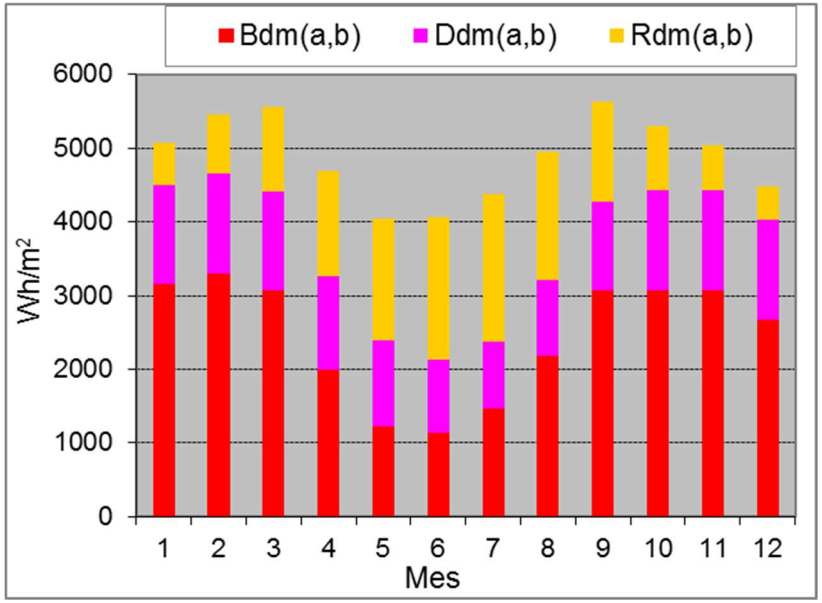
Inclinación, β (º)=	**90**
Orientación, α (º) =	**0**
Coeficiente ρ =	**0,5**

Ba(α,β) (kWh/m^2)	Da(α,β) (kWh/m^2)	Ra(α,β) (kWh/m^2)	**Ga(α,β)** (kWh/m^2)
893	**446**	**178**	**1.517**
59%	29%	12%	100%

Ba(α,β) (kWh/m^2)	Da(α,β) (kWh/m^2)	Ra(α,β) (kWh/m^2)	**Ga(α,β)** (kWh/m^2)
893	**446**	**445**	**1.784**
50%	25%	25%	100%

(+18%)





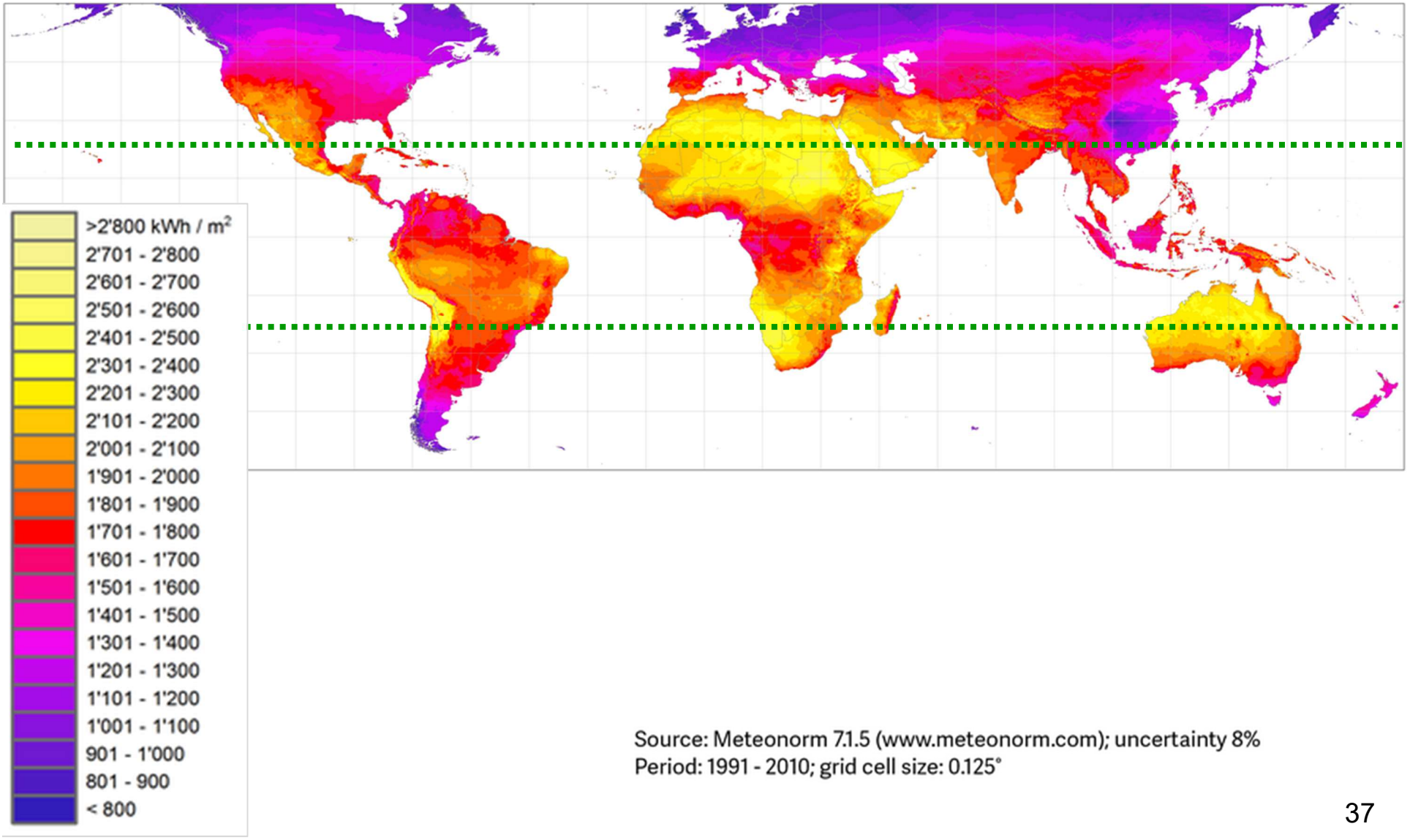
EJERCICIO PROPUESTO (FUERA DEL AULA)

Calcular, para la localidad en la que naciste, la irradiación anual máxima que podría captar una superficie, utilizando la hoja de cálculo de apoyo y los datos de partida proporcionados por alguna de las plataformas vistas en clase. Los datos deben ser promedios mensuales de irradiación diaria de, al menos, 10 años

¿Cómo habría que orientar e inclinar la superficie para captar ese máximo recurso solar)?

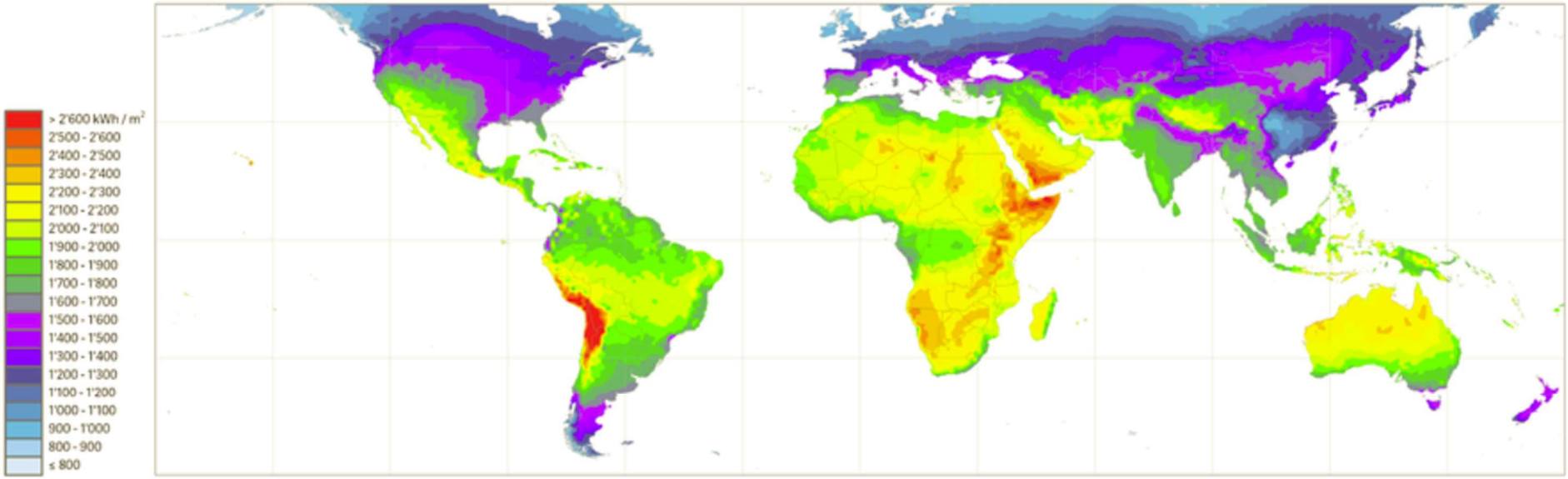
2. POTENCIAL SOLAR ANUAL DE SUPERFICIES

Yearly sum of Global Horizontal Irradiation (GHI)



Source: Meteonorm 7.1.5 (www.meteonorm.com); uncertainty 8%
Period: 1991 - 2010; grid cell size: 0.125°

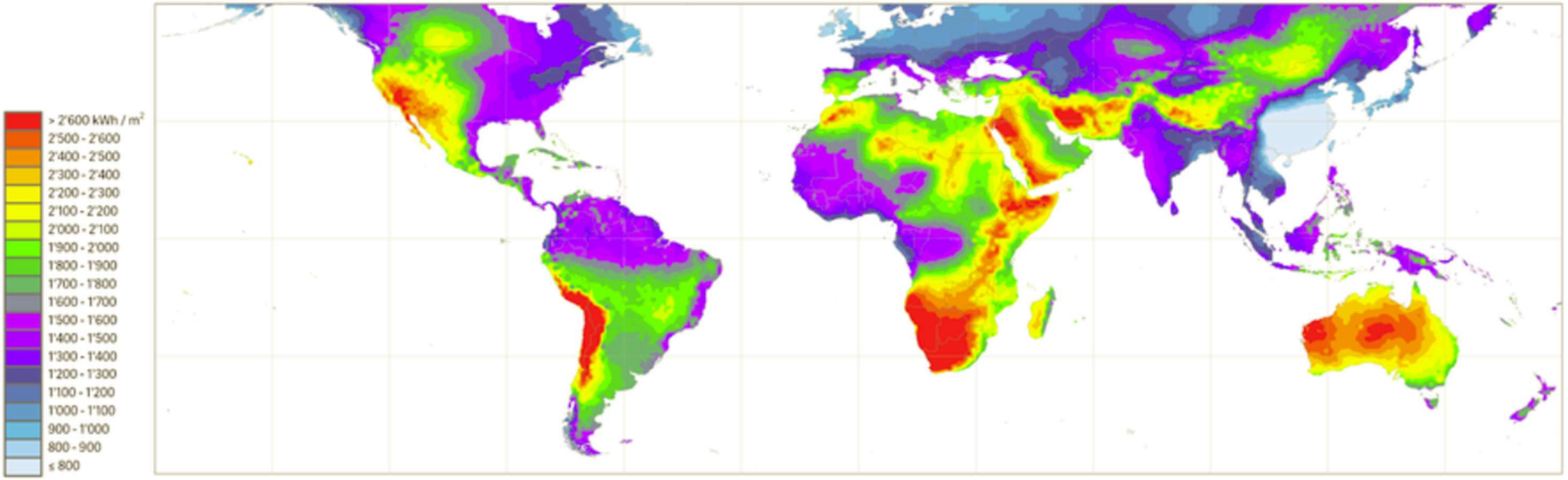
Yearly sum of Global Horizontal Irradiation (GHI)



Source: Meteonorm 8.1 (www.meteonorm.com); uncertainty 6%
Period: 1996-2015; grid cell size: 0.0625°

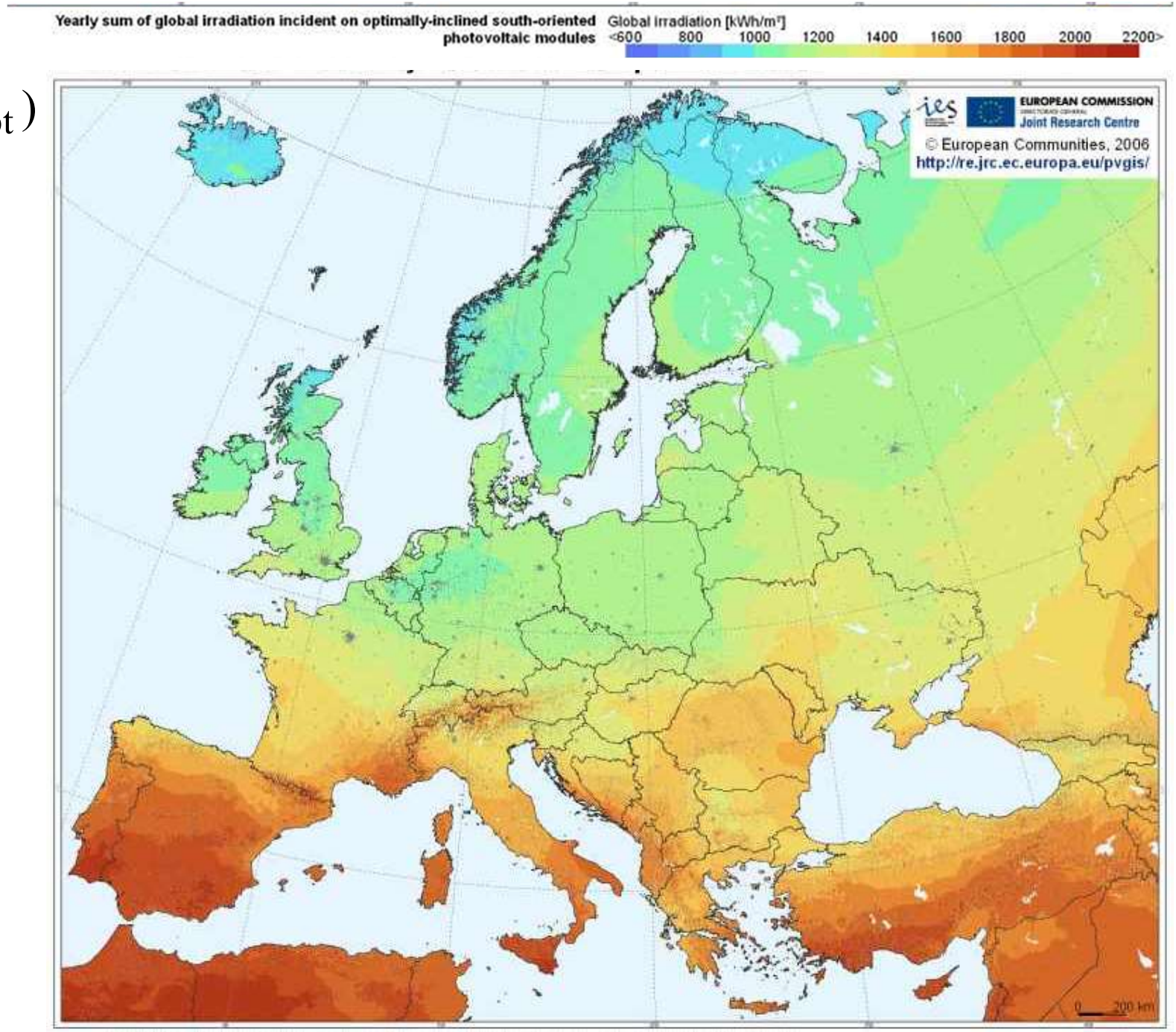
Meteonorm 8 Meteotest

Yearly sum of Direct Normal Irradiation (DNI)



Source: Meteonorm 8.1 (www.meteonorm.com); uncertainty 6%

$G_{\mathrm{a}}(\alpha_{\mathrm{opt}}, \beta_{\mathrm{opt}})$



El Año Meteorológico Típico (AMT) / Typical Meteorological Year (TMY)

- Año hipotético formado por valores reales de irradiación horaria horizontal y temperatura ambiente
- Elaborado a partir de bases de datos de varios años. Estadísticamente representativo de la climatología local
- Fuente de datos de gran interés para análisis del comportamiento energético anual sin perder la variabilidad natural
- Disponible en un número elevado de herramientas de simulación

Ejemplos:

$$365\ \text{días} \times \left\{ 24\ (\text{horas/día}) \times \left[G_{\text{h}}(0) + T_{\text{a,h}} \right] \right\}$$

$$365\ \text{días} \times \left\{ 24\ (\text{horas/día}) \times G_{\text{h}}(0) + \left(T_{\text{a,MAX}} + T_{\text{a,min}} \right)_{\text{dia}} \right\}$$

La "Superficie Óptima"

- Definición: Superficie fija que capta la máxima irradiación global anual
- Posición dependiente de los modelos de transposición y datos meteorológicos:

Orientación:

$$\alpha_{opt} = 0^{o}\,(\text{sur})\ \text{en hemisferio norte};\ 180^{o}\,(\text{norte})\ \text{en hemisferio sur}$$

Inclinación, primera aproximación:

$$\beta_{opt} \approx \phi\,(\text{latitud})\ \text{en cinturón intertropical}$$

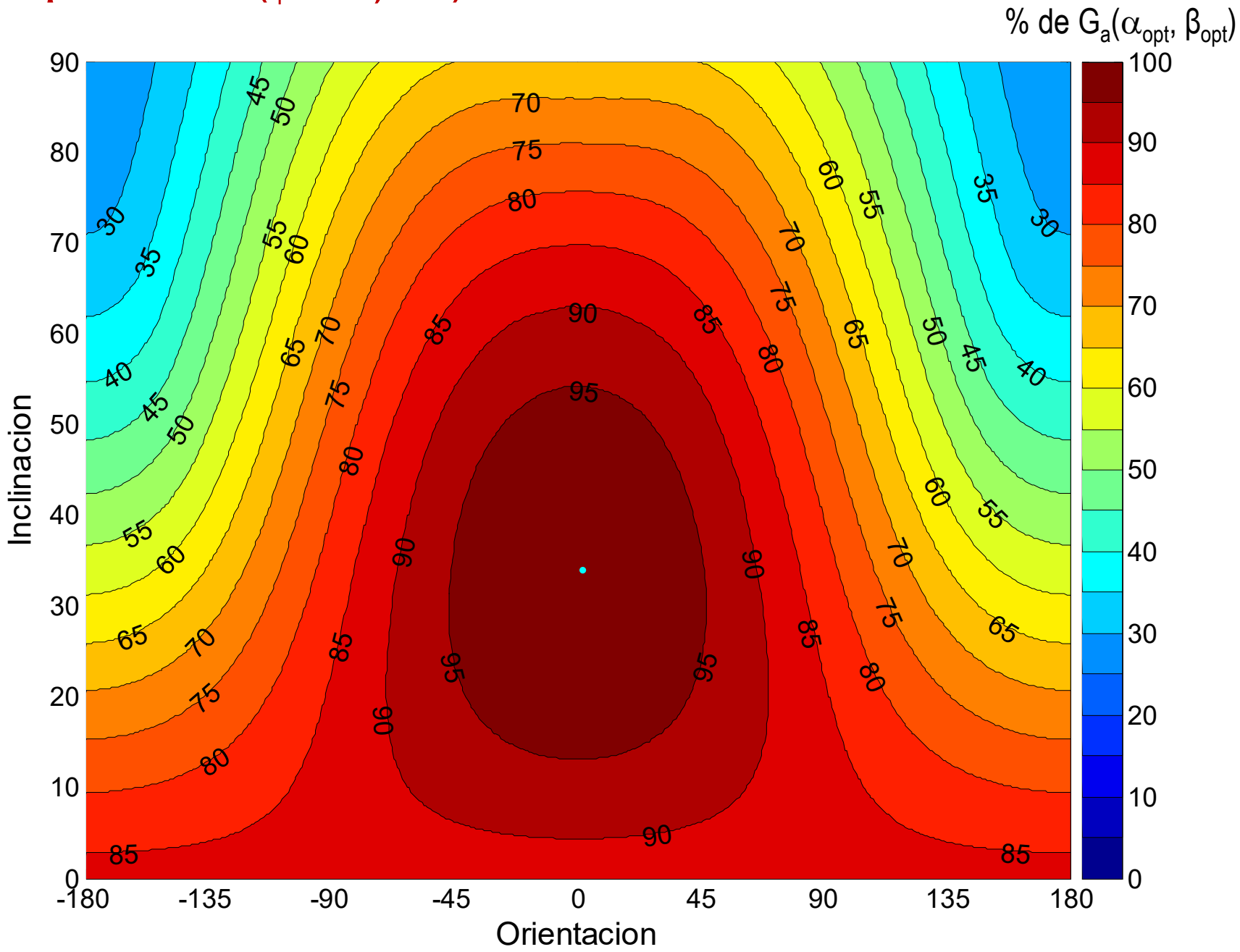
$$\approx \phi - 5^{o}\ \text{en latitudes medias}$$

$$\approx \phi - 10^{o}\ \text{en latitudes altas}$$

Cartas de Irradiación solar

- Representaciones gráficas que cuantifican el potencial solar de las superficies, relativo al máximo recurso solar de una localidad ($\leftrightarrow$ pérdidas con respecto de una superficie óptima)
- De gran utilidad en la fases iniciales de diseño del edificio (evaluación de envolventes en un edificio, adaptación a parcela existente)

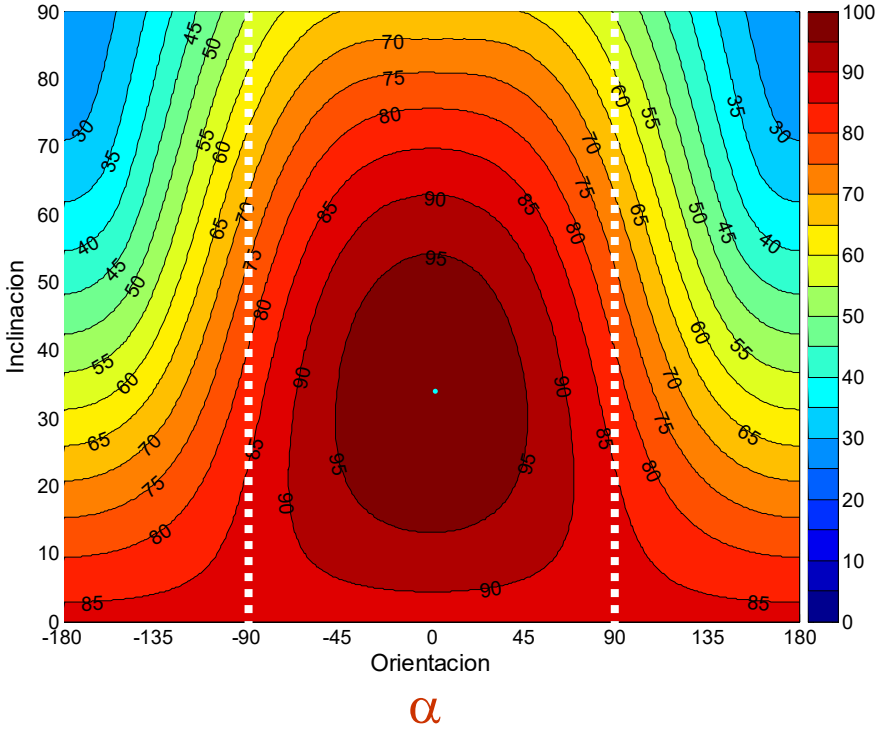
Ejemplo: Madrid ($\phi = 40{,}45°$)



Análisis de potencial solar: potencial relativo

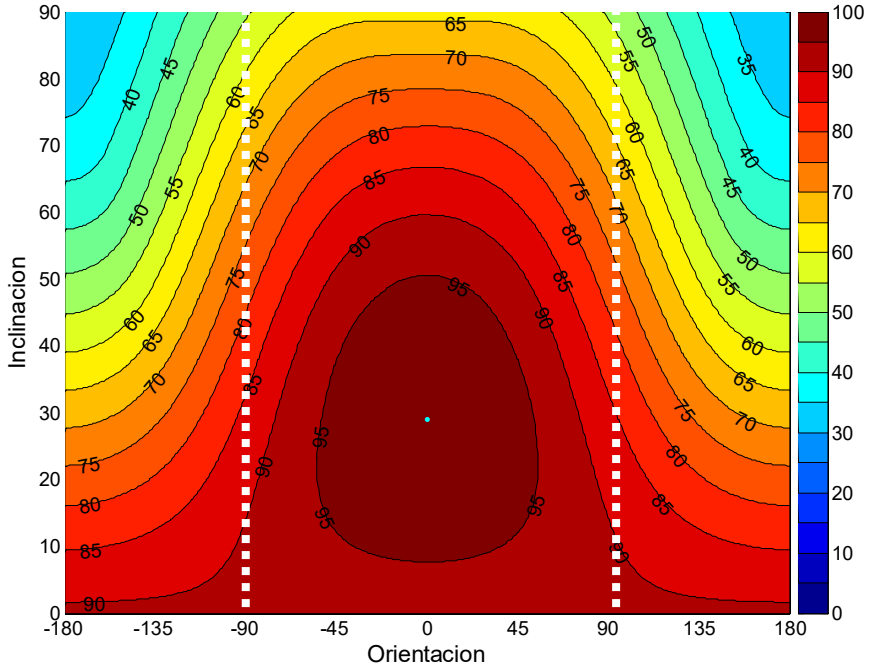
España, zonas climáticas CTE (2018):

Madrid (ϕ=40,5ºN, **zona IV CTE**)



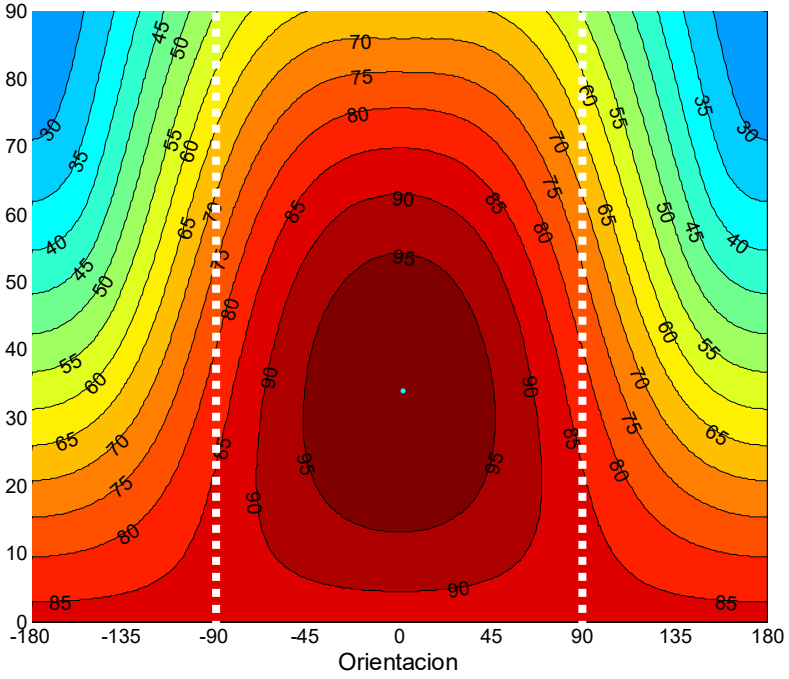
$G_a(\alpha_{opt}, \beta_{opt}) \sim$ **2200 kWh/m²**

Vitoria-Gasteiz (ϕ=42,8ºN, **zona I**)



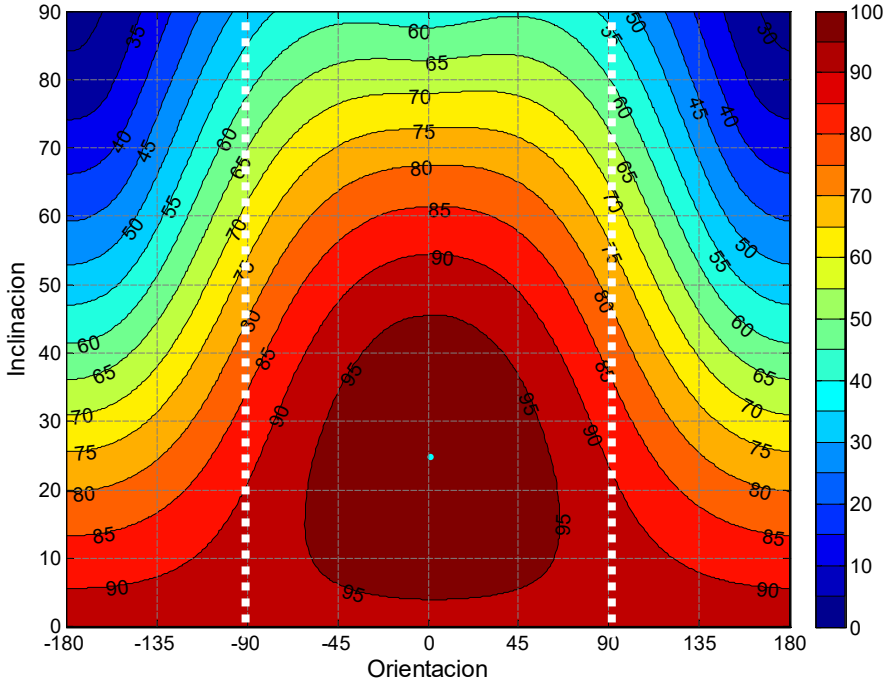
$G_a(\alpha_{opt}, \beta_{opt}) \sim$ **1700 kWh/m²**

Madrid (ϕ=40,5ºN, **zona IV**)



$G_a(\alpha_{opt}, \beta_{opt}) \sim$ **2200 kWh/m²**

Las Palmas (ϕ=28ºN, **zona V**)



$G_a(\alpha_{opt}, \beta_{opt}) \sim$ **2100 kWh/m²**

Potencial solar abundante en toda la geografía española